

Exact Identities for Polylogarithms, Harmonic Zeta Functions, and Stieltjes Finite Parts

Colliding grids, nonlinear coordinates, fully shifted sums,
and trilogarithmic antiderivatives

Research continuation prepared for Vladimir Reshetnikov
The ProveIt project

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Abstract

We continue the exact-identity program in the supplied ProveIt manuscript and its five current incoming polylogarithm reports. We obtain a holomorphic coefficient formula for every colliding unequal-dilation product of generalized Stieltjes functions and their argument derivatives, including a convergent collision expansion and an odd-order vanishing family. A finite Taylor-coefficient law determines every nonlinear coordinate correction, its composition rule, and its antiderivative endpoint constant. Three separated unequal frequencies reduce to a finite family of colored Tornheim kernels; complete local subtraction then gives all Stieltjes and argument derivatives. For fully shifted height-one harmonic sums, a Gamma-normalized difference is entire in the order, and all its nonpositive-integer values are finite Stirling transforms. This yields every Laurent principal part and finite part, with exact pole orders and Gamma antiderivatives. Finally, the weighted arctangent family left unevaluated in the canonical chapter is reduced to two values of one real trilogarithmic function, together with ordinary primitives and a double-polylogarithm integration ladder. Proofs distinguish coordinate finite parts from meromorphic coefficient extraction. The report records two pending source corrections, preserves the unresolved status of the short S_6 and current S_8 conjectures, and proposes precise next identity problems. Numerical and symbolic audits accompany, but do not replace, the proofs.

Source baseline. Repository commit `445f754610e1377939235794de84ed9075fc09f5`.

Status. Analytic research manuscript with executable audits; no Lean formalization or exhaustive historical-priority claim.

Contents

1	Scope, results, and proof status	3
1.1	The results at a glance	3
1.2	What the proofs have in common	4
1.3	Conjectures not settled by this report	4
2	Conventions and analytic preliminaries	4
2.1	Order derivatives, argument derivatives, and harmonic numbers	4
2.2	The coordinate finite part	5
2.3	Fourier normalization and the two-frequency kernel	6
3	Colliding unequal dilations: a complete finite-part calculus	6
3.1	The complete local subtraction polynomial	6
3.2	A holomorphic master kernel for every positive derivative order	7
3.3	Polygamma identities and a regularization diagnostic	10
3.4	The zero-derivative Stieltjes family	11
3.5	Exact matching with the noncollision expansion	12
3.6	Common dilation and an integration safeguard	13
4	Nonlinear endpoint coordinates: a finite exact law	14
4.1	A general coefficient lemma	14
4.2	Composition is an exact cocycle	15
4.3	Every Stieltjes index and every argument derivative	16
4.4	Exact antiderivatives and their endpoint constants	17
5	Three unequal frequencies and all their regularized jets	18
5.1	Weighted colored kernels and finite character filters	18
5.2	All Stieltjes and argument jets in the ambient coordinate	19
5.3	Independent polynomial checks	21
6	Fully shifted height-one sums: an entire difference and exact Laurent data	21
6.1	The common-shift convention	21
6.2	The exact entire difference	22
6.3	A bridge to independent inner and outer shifts	24
6.4	The complete Laurent principal part and finite part	25
6.5	The principal Stieltjes point	26
6.6	Explicit values and Gamma antiderivatives	27
6.7	Reproduction and further questions	27
7	A weighted arctangent integral in classical trilogarithms	28
7.1	The hyperbolic addition formula	28
7.2	The weighted inverse-hyperbolic companion	30
7.3	The classical logarithmic kernel	31
7.4	The ordinary primitive at a general endpoint	32
7.5	An elementary differential certificate	33
7.6	Every Euler integration order	33
7.7	An exact obstruction to the one-product ansatz	35
8	Source audit, reproducibility, and further research	35
8.1	Corrections and exact claim status	35
8.2	What the computational artifacts establish	37
8.3	Proposed exact research problems	37
	References	40

1 Scope, results, and proof status

This report addresses exact identities and the analytic mechanisms that make them valid. The source is the unified manuscript at `Analysis/Polylogarithms/docs/manuscript`, together with the five polylogarithm archives present in `docs/incoming` at the commit printed on the title page [1, 2, 3, 4, 5, 6]. All 85 retrieved file payloads were matched to their Git blob identifiers before the research comparison. The accompanying provenance manifest fixes precisely what was inspected; later changes to the repository are outside this comparison.

Several questions in those reports concern the point where an exact formula is known in a regular domain but its endpoint extension has not yet been justified. Typical examples are a collision of two rational singularity grids, replacing an affine endpoint coordinate by a nonlinear one, and taking all argument derivatives of a three-factor integral. We resolve these questions under explicit hypotheses. Another direction is a fully shifted nested harmonic family, where changing every denominator produces an entire correction that the outer-shift theory does not contain. A final direction returns to the canonical weighted arctangent integral and gives its complete real evaluation.

1.1 The results at a glance

1. **Every colliding two-grid Stieltjes product.** Theorem 3.1 gives a finite holomorphic generator for all Stieltjes indices, all argument derivative orders of positive total order, and all positive integer dilations. Section 3 also treats total order zero. The coefficients close in ordinary zeta derivatives, harmonic numbers, and logarithms. The regular constant in the convergent separated-shift collision expansion is exactly the independently defined colliding finite part. Corollary 3.3 gives an additional all-index odd-order cancellation.
2. **Every nonlinear endpoint correction.** Theorem 4.1 gives the complete change of a logarithmic-pole finite part from a finite jet of the coordinate. Theorems 4.3 and 4.4 give its composition and Stieltjes forms. A sharp disappearance threshold holds for coordinates tangent to the identity, and Theorem 4.6 gives exact endpoint constants for ordinary primitives evaluated by a coordinate finite part.
3. **Three unequal separated frequencies.** Theorem 5.1 gives a finite sign-cone and character formula involving $2(p_1 + p_2 + p_3)$ weighted colored Tornheim kernels. Each weighted kernel is itself a finite sum of ordinary colored kernels. Theorem 5.2 performs complete ambient-coordinate subtraction for every Stieltjes index and argument derivative.
4. **An entire difference for fully shifted harmonic sums.** Theorem 6.1 isolates a universal Gamma-normalized singular part and evaluates the entire remainder at every nonpositive integer by a finite Stirling sum. Theorem 6.3 gives the complete Laurent principal part and finite part at all depths. The pole order drops at negative even integers for a proved Bernoulli reason. A two-shift bridge and explicit Gamma antiderivatives follow.
5. **Weighted arctangent and inverse-hyperbolic identities.** Theorem 7.1 evaluates the canonical weighted family by a hyperbolic addition formula for one even trilogarithmic function. Theorem 7.4 supplies a normalized ordinary primitive at every endpoint in $(0, \pi)$, with principal branches justified. Theorem 7.6 gives every Euler integration order in depth-two polylogarithms. A trigonometric companion evaluates a weighted inverse-hyperbolic family, including its endpoint value.

These are proved extensions relative to the inspected source material. They are not statements that every displayed identity is historically new. In particular, the Hurwitz Fourier product, Gamma multiplication, Gauss connection formula, and elementary logarithmic primitives are classical inputs [7, 8, 12, 15, 18]. The depth-one harmonic Hurwitz-zeta literature already treats continuation and Laurent coefficients [10, 11]; the limited access to the newest preprint is disclosed in Section 6.

1.2 What the proofs have in common

A local term of the form $c(z)t^{z-1}$ explains much of the endpoint calculus. Its meromorphic integral over $(0, h)$ is $c(z)h^z/z$. The generating function of its coefficientwise, fixed-coordinate finite parts is instead

$$\frac{c(z)(h^z - 1)}{z}.$$

The difference is the complete expression $c(z)/z$. Retaining only its residue leaves incorrect regular coefficients whenever c depends on z . Nonlinear coordinates, dilations, and Taylor coefficients of other factors all produce precisely this dependence. The local subtraction proofs below retain it before taking any coefficient.

The fully shifted harmonic theorem uses the complementary mechanism. Its Gauss connection formula splits the Mellin kernel into a universal singular block and an analytic block. Division by $\Gamma(s)$ cancels the analytic block's simple Mellin poles. At $s = -m$ the resulting entire function is determined by one Taylor coefficient, which turns out to be a finite Stirling polynomial. Thus exact local data control both the distributional correction and the entire harmonic correction, although the two operations are analytically different.

1.3 Conjectures not settled by this report

The source's S_4 identity is already proved by its octahedral rational certificate. The short S_6 candidate and the current, replacement S_8 candidate remain conjectures. The rejected earlier S_8 vector remains rejected. The numerically supported higher golden ladder vectors are also not proved here. The present exact results do not imply those reductions, and their numerical evidence has not been used as a proof. Section 8 records exact source labels and the integration consequences.

2 Conventions and analytic preliminaries

2.1 Order derivatives, argument derivatives, and harmonic numbers

For $\Re a > 0$, the Hurwitz zeta function is initially

$$\zeta(s, a) = \sum_{n=0}^{\infty} (n+a)^{-s}, \quad \Re s > 1,$$

with the principal logarithm, and then has its usual meromorphic continuation. We use

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n, \quad \gamma_n = \gamma_n(1), \quad \gamma_0 = \gamma. \quad (2.1)$$

Thus the generator near $z = 0$ is

$$\mathcal{G}(z, a) = \zeta(1-z, a) + \frac{1}{z} = \sum_{n \geq 0} \gamma_n(a) \frac{z^n}{n!}, \quad \gamma_0(a) = -\psi(a). \quad (2.2)$$

A superscript on ζ , as in $\zeta^{(j)}(s, a)$, means ∂_s^j ; a superscript on γ_n or ψ means an argument derivative. The notation D_y is used for a distributional derivative. These three operations must be distinguished. The standard identities used below are

$$\zeta(s, a) = a^{-s} + \zeta(s, a+1), \quad (2.3)$$

$$\partial_a^r \zeta(s, a) = (-1)^r (s)_r \zeta(s+r, a), \quad (2.4)$$

$$\psi^{(r)}(a) = (-1)^{r+1} r! \zeta(r+1, a) \quad (r \geq 1). \quad (2.5)$$

They fix the normalizations and agree with DLMF [12, 13].

For later coefficient algebra put

$$H_N^{(s)}(a) = \sum_{j=0}^{N-1} (a+j)^{-s} = \zeta(s, a) - \zeta(s, N+a), \quad H_N^{(s)} = H_N^{(s)}(1).$$

This finite sum is entire in s ; every order derivative is obtained by inserting $(-\log(a+j))^k$. For integer $s \geq 2$, it is also a difference of polygamma values by (2.5). We set $H_0^{(s)} = 0$. The elementary symmetric functions of $1, 1/2, \dots, 1/r$ are finite polynomials in $H_r, H_r^{(2)}, \dots$ by Newton's identities. Coefficient extraction $[z^n t^j]$ always denotes the ordinary Taylor coefficient; factorials are printed explicitly.

For example, (2.4) gives the useful classical bridge

$$\gamma_n^{(r)}(a) = (-1)^{n+r} r! \sum_{h=0}^{\min(n,r)} \frac{n!}{(n-h)!} e_h(1, 1/2, \dots, 1/r) \zeta^{(n-h)}(r+1, a), \quad r \geq 1. \quad (2.6)$$

Indeed, take the coefficient of z^n in $(-1)^r (1-z)_r \zeta(r+1-z, a)$. An ordinary primitive is

$$\int \gamma_n(a) da = \frac{(-1)^{n+1}}{n+1} \zeta^{(n+1)}(0, a) + C. \quad (2.7)$$

One proof takes coefficients in $\partial_a \zeta(-z, a) = z \zeta(1-z, a)$ after removing the constant spectral pole. At $n=0$ this agrees with $-\log \Gamma(a) + C$. These identities are baseline transformations, not claims of new results; the endpoint constants after nonlinear substitution require the additional analysis of Section 4.

2.2 The coordinate finite part

Suppose a function has finitely many singular points and, to the right of each point, a finite sum of nonintegrable terms $t^{-q}(\log t)^n$ plus a locally integrable remainder. Integrate outside right intervals of lengths ε_j and expand in these cutoffs. The symbol FP_x means the coefficient of $\prod_j \varepsilon_j^0 (\log \varepsilon_j)^0$ in this expansion, with $t = x - x_j$ the original coordinate. Independent cutoffs are permitted. A common cutoff gives the same constant because the finitely many endpoint contributions add. No left cutoff is needed for a fractional-part Hurwitz singularity: its left limit is the regular branch at argument one.

For distributions the same prescription is applied after pairing with a smooth test function and subtracting its Taylor polynomial. It is local and linear. Products in this report are finite parts of *pointwise products* with explicitly subtracted local expansions. No undefined multiplication of colliding singular distributions is invoked. In contrast, a coefficient of a meromorphic family is a spectral operation; the two operations coincide only after the required complete local pole terms have been accounted for.

The periodic Hurwitz distribution $Z(1-z)$ is defined by the ordinary locally integrable function for $\Re z > 0$ and meromorphically continued by local Taylor subtraction. At an integer, in a signed local coordinate y , its decomposition is

$$Z(1-z) = y_+^{z-1} + \zeta(1-z, 1+y).$$

Consequently

$$Z(1-z) = \frac{\delta_0 - 1}{z} + \sum_{n \geq 0} G_n \frac{z^n}{n!}, \quad G_n = \text{FP}_y \gamma_n(\{y\}). \quad (2.8)$$

Here 1 means the constant periodic distribution. This construction also fixes the normalization of every $D_y^r G_n$ used below.

2.3 Fourier normalization and the two-frequency kernel

Write $e(t) = e^{2\pi it}$. For $\Re z > 1$, the absolutely convergent Hurwitz Fourier series in our convention is

$$\zeta(1-z, x) = \frac{\Gamma(z)}{(2\pi)^z} \sum_{n \neq 0} \frac{e^{-i\pi z \operatorname{sgn}(n)/2}}{|n|^z} e(nx). \quad (2.9)$$

For positive integers p, q set $d = \gcd(p, q)$, $P = p/d$, $Q = q/d$, and $\tau = Qa - Pb$. Orthogonality leaves frequencies $Q\ell, -P\ell$ and therefore gives

$$\begin{aligned} C(z, w; p, q; a, b) &:= \int_0^1 \zeta(1-z, \{px+a\}) \zeta(1-w, \{qx+b\}) dx \\ &= \frac{\Gamma(z)\Gamma(w)Q^{-z}P^{-w}}{(2\pi)^{z+w}} \left\{ e^{-i\pi(z-w)/2} \operatorname{Li}_{z+w}(e(\tau)) + e^{i\pi(z-w)/2} \operatorname{Li}_{z+w}(e(-\tau)) \right\}. \end{aligned} \quad (2.10)$$

Initially $\Re z, \Re w > 1$; subsequently the formula means meromorphic continuation. For $\tau \in \mathbb{Z}$ the two polylogarithms are $\zeta(z+w)$. The unit-frequency coincident case is the classical Espinosa–Moll product [7, Theorem 3.1]. The shifted and dilated versions in the incoming reports have this normalization. Formula (2.10) makes the pair kernels needed in Section 5 explicit within this report.

The Riemann and Hurwitz functional equations, Gamma reflection and multiplication, and polylogarithm order continuation are used in their standard forms [12, 17]. Each use of a singular parameter value below is licensed by a local continuation proof before its Taylor or Laurent coefficient is taken.

3 Colliding unequal dilations: a complete finite-part calculus

The separated-grid theorem in the incoming Mellin–dilation report [3] explicitly excludes $Pa \in \mathbb{Z}$. At these shifts its partition of unity no longer defines a product of the two singular distributions. This section treats precisely that omitted case. The construction is the constant term of the pointwise product in a fixed ambient coordinate; it neither assumes that a product of colliding distributions exists nor identifies a bare spectral Laurent coefficient with that constant term.

Let p, q be positive integers and put

$$d = \gcd(p, q), \quad P = p/d, \quad Q = q/d, \quad A = \log P, \quad B = \log Q, \quad K = \log \operatorname{lcm}(p, q).$$

For $m, n, r, k \geq 0$, write $N = r + k$ and define

$$J_{mn;r,k}^{p,q} = \operatorname{FP}_x \int_0^1 \gamma_m^{(r)}(\{px\}) \gamma_n^{(k)}(\{qx\}) dx. \quad (3.1)$$

The derivatives are in the special-function argument. At every member of the union of the two singular grids, delete the right interval of ambient length ϵ and retain the constant term as $\epsilon \downarrow 0$. Equivalently, take separate constant terms in independent ambient cutoff variables at the finitely many points.

Every colliding shift a gives the same value as (3.1). Indeed, $Pa = h \in \mathbb{Z}$, and a solution of $Qj \equiv -h \pmod{P}$ makes $x_0 = j/p$ satisfy $px_0, qx_0 + a \in \mathbb{Z}$. Translation by x_0 carries the two fractional-part arguments to px and qx and preserves ambient lengths. Thus no collision cases are lost by writing the common shift as zero.

3.1 The complete local subtraction polynomial

Use the holomorphic Stieltjes generating function

$$\mathcal{G}(z, x) = \zeta(1-z, x) + \frac{1}{z} = \sum_{m \geq 0} \gamma_m(x) \frac{z^m}{m!}, \quad p_r(z) = \frac{(1-z)_r}{r!} = \prod_{j=1}^r \left(1 - \frac{z}{j}\right).$$

For $r = 0$ the empty product is one. Set

$$S_{m,r}(L) = m![z^m] p_r(z) e^{zL} = \sum_{j=0}^{\min(m,r)} (-1)^j \frac{m!}{(m-j)!} e_j(1, 1/2, \dots, 1/r) L^{m-j}. \quad (3.2)$$

The Hurwitz shift identity gives the exact local decomposition

$$\gamma_m^{(r)}(y) = (-1)^r r! y^{-r-1} S_{m,r}(\log y) + \gamma_m^{(r)}(1+y). \quad (3.3)$$

Here is an explicit subtraction rule that covers every grid configuration. At a point b of the first grid put $t = x - b > 0$ and

$$U_{p,m,r}(t) = (-1)^r r! p^{-r-1} t^{-r-1} S_{m,r}(\log p + \log t).$$

If b is not on the second grid, subtract

$$U_{p,m,r}(t) \sum_{j=0}^r \frac{q^j t^j}{j!} \gamma_n^{(k+j)}(\{qb\}).$$

At a point of the second grid only, use the analogous expression with the two factors exchanged. At a common point subtract the sum of

$$U_{p,m,r}(t) U_{q,n,k}(t), \quad (3.4)$$

$$U_{p,m,r}(t) \sum_{j=0}^r \frac{q^j t^j}{j!} \gamma_n^{(k+j)}(1), \quad (3.5)$$

$$U_{q,n,k}(t) \sum_{j=0}^k \frac{p^j t^j}{j!} \gamma_m^{(r+j)}(1). \quad (3.6)$$

The first line has power t^{-N-2} and a polynomial in $\log t$. The other lines contain the powers $t^{-r-1}, \dots, t^{-1}$ and $t^{-k-1}, \dots, t^{-1}$. These are all nonintegrable terms.

Taylor's theorem and (3.3) show that the remaining integrand is $O(1 + |\log t|^{m+n})$. It is therefore integrable. Adding the elementary finite-part primitives of the subtracted terms proves existence of (3.1). For clarity, the primitive normalization is

$$\text{FP} \int_0^\ell t^{-1} \log^j t \, dt = \frac{\log^{j+1} \ell}{j+1},$$

and, for $h \geq 2$,

$$\text{FP} \int_0^\ell t^{-h} \log^j t \, dt = \left. \partial_\alpha^j \frac{\ell^{\alpha-h+1}}{\alpha-h+1} \right|_{\alpha=0}.$$

No finite local constant is omitted or added.

3.2 A holomorphic master kernel for every positive derivative order

For $M \geq 1$ define

$$W_M(z) = (1-z)_M \zeta(M+1-z), \quad (3.7)$$

$$R(z, w) = \frac{\Gamma(1-z-w) \Gamma(1+z)}{\Gamma(1-w)}, \quad (3.8)$$

$$V_M(z, w) = R(z, w) W_M(z+w) = \frac{\Gamma(M+1-z-w) \Gamma(1+z)}{\Gamma(1-w)} \zeta(M+1-z-w). \quad (3.9)$$

In particular $V_M(0, w) = W_M(w)$. All these functions are holomorphic in a sufficiently small polydisc about $(0, 0)$.

Theorem 3.1 (Colliding-grid master theorem). *For $N = r + k \geq 1$, the exponential generating function of the ambient-coordinate finite parts is*

$$\begin{aligned} \mathcal{J}_{r,k}^{p,q}(z, w) &:= \sum_{m,n \geq 0} J_{mn;r,k}^{p,q} \frac{z^m w^n}{m!n!} \\ &= Q^r P^k e^{-Bz-Aw} \left\{ (-1)^k \frac{V_N(z, w) - e^{Kz} p_r(z) W_N(w)}{z} \right. \\ &\quad \left. + (-1)^r \frac{V_N(w, z) - e^{Kw} p_k(w) W_N(z)}{w} \right\}. \end{aligned} \quad (3.10)$$

Both displayed divided differences have removable denominators separately. Consequently (3.10) is a finite coefficient algorithm for every m, n, r, k, p, q .

Proof. First work where all derivatives of the Hurwitz factors have absolutely convergent Fourier series. Orthogonality leaves the frequency indices $Q\ell$ and $-P\ell$. The classical coincident Hurwitz product formula [7, Theorem 3.1], or the same Fourier calculation followed by the Riemann functional equation, gives

$$\begin{aligned} \mathcal{C}_{r,k}^{p,q}(z, w) &:= \int_0^1 \partial_y^r \zeta(1-z, \{px\}) \partial_y^k \zeta(1-w, \{qx\}) dx \\ &= Q^r P^k e^{-Bz-Aw} \left\{ (-1)^k \frac{V_N(z, w)}{z} + (-1)^r \frac{V_N(w, z)}{w} \right\}. \end{aligned} \quad (3.11)$$

The integral notation on the left subsequently means meromorphic continuation. One can verify the signs directly from $\partial_y^r \zeta(s, y) = (-1)^r (s)_r \zeta(s+r, y)$ and $(-1)^r (s)_r \Gamma(1-s-r) = \Gamma(1-s)$.

The local subtraction above continues the integral and identifies its Taylor coefficients with fixed-coordinate finite parts. We detail the terms that distinguish these coefficients from those of (3.11). At a first-grid point, the singular generating factor is

$$(-1)^r r! p^{-r-1} p_r(z) e^{z \log p} t^{z-r-1}.$$

Only the Taylor coefficient of degree r of the other factor creates a pole at $z = 0$. At a common point the other factor is its regular piece at argument 1; at a noncommon point it is its ordinary value. These arguments, over the first grid, form d copies of $1/P, 2/P, \dots, 1$. Since $N \geq 1$, their trace is

$$\frac{1}{p} \sum_{b \text{ in first grid}} \partial_y^N \mathcal{G}(w, \eta_b) = (-1)^N P^{N-w} W_N(w), \quad \eta_b = \begin{cases} \{qb\}, & \{qb\} > 0, \\ 1, & \{qb\} = 0. \end{cases}$$

This is precisely the Hurwitz multiplication formula differentiated N times in the argument.

The complete t^{z-1} coefficient, summed over the first grid, is

$$c_z(z, w) = (-1)^k Q^r P^k e^{z \log p - Aw} p_r(z) W_N(w).$$

The meromorphic integral of $c_z t^{z-1}$ on $(0, \ell)$ is $c_z \ell^z / z$. The generating function of its coefficientwise ambient finite parts is instead $c_z(\ell^z - 1)/z$. Thus the correction is the *complete* quantity $-c_z(z, w)/z$, not merely its residue at $z = 0$. The second grid contributes

$$-\frac{c_w(z, w)}{w}, \quad c_w(z, w) = (-1)^r Q^r P^k e^{w \log q - Bz} p_k(w) W_N(z).$$

At common points the singular-singular term is proportional to $t^{z+w-N-2}$. Its integral has denominator $z + w - N - 1$, which is nonzero near the origin. Its analytic integral already equals the generating function of its finite parts; there is no additional diagonal contact term at $(0, 0)$.

All remaining subtracted integrals converge locally uniformly on a small spectral polydisc. Indeed the Taylor remainders gain enough powers of t , and their spectral derivatives introduce only powers

of $\log t$. They are holomorphic by dominated differentiation. Subtracting $c_z/z + c_w/w$ from (3.11) proves (3.10), since $\log p + B = \log q + A = K$. If one of r, k is zero, replacing that spectral factor by $\mathcal{G}$ adds its scalar $1/z$ or $1/w$; the integral of the other, positive-order factor is zero by Fourier orthogonality and meromorphic continuation. Hence no extra scalar term occurs when $N \geq 1$. $\square$

The proof isolates a useful fact: common points introduce a stronger power divergence, but the finite discrepancy between spectral and coordinate regularization is governed by the resonant t^{-1} terms. The stronger merged term must be subtracted, yet its denominator has no spectral pole at the parameter values being differentiated.

The comparison with rectangular spectral coefficient extraction is completely explicit. Define

$$\kappa_{m,r}(L) = m![z^{m+1}]p_r(z)e^{Lz}, \quad \omega_{n,N}(L) = n![w^n]e^{Lw}W_N(w),$$

and let $\text{LC}_{mn} \mathcal{C}$ denote $m!n![z^m w^n]$ in the rectangular Laurent expansion of (3.11). Then

$$\begin{aligned} J_{mn;r,k}^{p,q} - \text{LC}_{mn} \mathcal{C}_{r,k}^{p,q} = & -Q^r P^k \{(-1)^k \kappa_{m,r}(\log p) \omega_{n,N}(-A) \\ & + (-1)^r \kappa_{n,k}(\log q) \omega_{m,N}(-B)\}. \end{aligned} \quad (3.12)$$

This finite formula follows by extracting the two complete local corrections in the proof. It specifies the difference between the two regularizations for every index, not only for the zero-index examples.

Corollary 3.2 (Finite closure in ordinary zeta derivatives). *For $N \geq 1$, every $J_{mn;r,k}^{p,q}$ is a finite linear combination of $\zeta^{(j)}(N+1)$, $0 \leq j \leq m+n+1$. Its coefficients lie in the $\mathbb{Q}$ -algebra generated by A, B, K , the rational generalized harmonic numbers, and positive integer zeta values. No unevaluated multiple sum is required. Writing $M = m+n$, the coefficient of the highest derivative $\zeta^{(M+1)}(N+1)$ is*

$$(-1)^{M+1} N! Q^r P^k \left(\frac{(-1)^k}{m+1} + \frac{(-1)^r}{n+1} \right). \quad (3.13)$$

Proof. The finite products p_r and $(1-z)_N$ have rational coefficients. Moreover

$$\log R(z, w) = \sum_{j \geq 2} \frac{\zeta(j)}{j} ((z+w)^j + (-1)^j z^j - w^j). \quad (3.14)$$

Only total degree at most $m+n+1$ can enter a coefficient of (3.10). Taylor expansion of $\zeta(N+1-z-w)$ therefore uses derivatives of order at most $m+n+1$. The vanishing constant numerator makes division by z or w an exact shift of polynomial coefficients. The highest derivative can arise only from the degree- $M+1$ term of $\zeta(N+1-z-w)$; the two binomial coefficients after division by z and w give (3.13). The contact products involve derivatives of order at most n or m , respectively, and cannot alter this coefficient. $\square$

There is a stronger cancellation for equal Stieltjes indices after the elementary logarithmic shifts forced by the reduced dilations. Define

$$\gamma_{m,L}^{(r)}(x) = \sum_{j=0}^m \binom{m}{j} L^{m-j} \gamma_j^{(r)}(x).$$

Its generating function is $e^{Lz} \partial_x^r \mathcal{G}(z, x)$.

Corollary 3.3 (Odd-order diagonal collapse). *If $r+k = N$ is odd, then for every $m \geq 0$,*

$$\begin{aligned} \text{FP}_x \int_0^1 \gamma_{m,B}^{(r)}(\{px\}) \gamma_{m,A}^{(k)}(\{qx\}) dx \\ = (-1)^k Q^r P^k (\kappa_{m,k}(K) - \kappa_{m,r}(K)) \omega_{m,N}(0). \end{aligned} \quad (3.15)$$

Thus only zeta derivatives of order at most m remain, whereas a general pair with these indices can require order $2m + 1$. In particular,

$$\text{FP} \int_0^1 \gamma_m^{(r)}(x) \gamma_m^{(k)}(x) dx = 0, \quad r + k \text{ odd}, \quad m \geq \max(r, k). \quad (3.16)$$

For $r = 0, k = 1$ the unequal-dilation identity becomes, for $m \geq 1$,

$$\begin{aligned} \text{FP}_x \int_0^1 \gamma_{m,B}(\{px\}) \gamma'_{m,A}(\{qx\}) dx \\ = P(-K)^m \{ \zeta^{(m)}(2) + m \zeta^{(m-1)}(2) \}. \end{aligned} \quad (3.17)$$

Its $m = 0$ value is $P\zeta(2)$.

Proof. Multiplying (3.10) by e^{Bz+Aw} removes its external exponential. Since $(-1)^r = -(-1)^k$, the two spectral terms $V_N(z, w)/z$ and $V_N(w, z)/w$ have opposite diagonal coefficients. Only the two explicit contact products remain, giving (3.15). For $p = q = 1$, one has $K = 0$; $\kappa_{m,r}(0) = 0$ for $m \geq r$ because p_r has degree r . This proves (3.16). Finally $p_1(z) - p_0(z) = -z$ gives $\kappa_{m,1}(K) - \kappa_{m,0}(K) = -K^m$, and

$$\omega_{m,1}(0) = (-1)^m \{ \zeta^{(m)}(2) + m \zeta^{(m-1)}(2) \}, \quad m \geq 1.$$

These two identities prove the last formula. $\square$

3.3 Polygamma identities and a regularization diagnostic

Since $\gamma_0 = -\psi$, the zero-index specialization is particularly short. Put $H_0 = 0$ and $H_j = \sum_{h=1}^j 1/h$.

Corollary 3.4 (Every coincident polygamma product). *For $N = r + k \geq 1$,*

$$\begin{aligned} \text{FP}_x \int_0^1 \psi^{(r)}(\{px\}) \psi^{(k)}(\{qx\}) dx \\ = N! Q^r P^k \left\{ (-1)^{k+1} [\zeta'(N+1) + (H_N - H_r + K) \zeta(N+1)] \right. \\ \left. + (-1)^{r+1} [\zeta'(N+1) + (H_N - H_k + K) \zeta(N+1)] \right\}. \end{aligned} \quad (3.18)$$

In particular, for odd N ,

$$\text{FP}_x \int_0^1 \psi^{(r)}(\{px\}) \psi^{(k)}(\{qx\}) dx = (-1)^{k+1} N! Q^r P^k (H_k - H_r) \zeta(N+1). \quad (3.19)$$

These odd-order values are rational multiples of π^{N+1} and are unchanged by a common integer dilation of p, q .

Proof. At the origin $W_N(0) = N! \zeta(N+1)$ and

$$\partial_z V_N(0, 0) = -N! \{ \zeta'(N+1) + H_N \zeta(N+1) \}, \quad p'_r(0) = -H_r.$$

Take the two removable limits in (3.10). If N is odd, r, k have opposite parity, so the spectral derivatives and all scale terms cancel. Euler's formula for even zeta values gives the stated rational multiples of π^{N+1} . $\square$

For example,

$$\text{FP}_x \int_0^1 \psi(\{2x\}) \psi'(\{3x\}) dx = \frac{\pi^2}{3}, \quad (3.20)$$

$$\text{FP}_x \int_0^1 \psi'(\{2x\}) \psi'(\{3x\}) dx = 12 \{ 2\zeta'(3) + (1 + 2 \log 6) \zeta(3) \}, \quad (3.21)$$

$$\text{FP}_x \int_0^1 \psi''(\{2x\}) \psi'(\{3x\}) dx = -54 \zeta(4) = -\frac{3\pi^4}{5}. \quad (3.22)$$

The spectral continuation alone gives a different prescription. Already for $p = q = 1, r = 0, k = 1$, its rectangular Laurent constant is zero, while

$$\text{FP} \int_0^1 \psi(x) \psi'(x) dx = \zeta(2).$$

This can be checked without special-function product theory: $\int_\epsilon^1 \psi \psi' = \frac{1}{2}(\psi(1)^2 - \psi(\epsilon)^2)$, and $\psi(\epsilon) = -1/\epsilon - \gamma + \zeta(2)\epsilon + O(\epsilon^2)$. The constant is $\zeta(2)$. More generally, substituting $z = \alpha t, w = t$ into that bare spectral kernel gives constant $-(\zeta(2) + \zeta'(2))(\alpha - \alpha^{-1})$. Thus an unspecified one-parameter spectral finite part is not the ambient-coordinate operation. The local corrections in (3.10) remove this ambiguity before any coefficient is extracted.

Two first-index identities illustrate the Stieltjes extension. With the same P, A, K as above,

$$J_{10;0,1}^{p,q} = P \left\{ \frac{\zeta''(2)}{2} + K\zeta'(2) + \left(\frac{K^2}{2} + \log p - 1 \right) \zeta(2) \right\}, \quad (3.23)$$

$$J_{01;0,1}^{p,q} = P \left\{ -\frac{\zeta''(2)}{2} - (K+1)\zeta'(2) - \left(\frac{K^2}{2} + A \right) \zeta(2) \right\}. \quad (3.24)$$

Their sum for $p = q = 1$ is $-\zeta'(2) - \zeta(2)$, in agreement with the direct endpoint finite part of $(\gamma_1 \gamma_0)'$.

3.4 The zero-derivative Stieltjes family

For $r = k = 0$, the positive-order kernel must be supplemented by scalar spectral-pole terms. It is clearer to give a holomorphic formula using the undiluted coincident kernel, made explicit here:

$$\mathcal{J}_0(z, w) = \sum_{m,n \geq 0} J_{mn;0,0}^{1,1} \frac{z^m w^n}{m!n!}, \quad g(z) = \zeta(1-z) + \frac{1}{z} = \sum_{m \geq 0} \gamma_m \frac{z^m}{m!}.$$

The classical coincident spectral kernel is

$$\mathcal{C}_{0,0}^{1,1}(z, w) = 2\Gamma(z)\Gamma(w)(2\pi)^{-z-w} \cos\left(\frac{\pi(z-w)}{2}\right) \zeta(z+w). \quad (3.25)$$

The same direct endpoint subtraction gives

$$\mathcal{J}_0(z, w) = \mathcal{C}_{0,0}^{1,1}(z, w) + \frac{1}{zw} - \frac{g(w)}{z} - \frac{g(z)}{w}. \quad (3.26)$$

This grouped expression is holomorphic, as one can also see entirely algebraically. Put

$$T(z, w) = \frac{wR(z, w) + zR(w, z)}{z+w}.$$

The numerator vanishes at $w = -z$, because $R(z, -z) = R(-z, z) = 1$. Thus T is holomorphic, and $T(z, 0) = T(0, w) = 1$. Applying the Riemann functional equation to (3.25) gives

$$\mathcal{J}_0(z, w) = \frac{1 - T(z, w) + (z+w)T(z, w)g(z+w) - wg(w) - zg(z)}{zw}. \quad (3.27)$$

The numerator vanishes on both coordinate axes, so division by zw is removable. This provides a self-contained finite coefficient rule for the undiluted zero-derivative case, without a previously tabulated coefficient array.

Write $L_p = \log p$ and $L_q = \log q$. Direct local subtraction, exactly as in the master theorem, gives

$$\begin{aligned} \mathcal{J}_{0,0}^{p,q}(z, w) &= e^{-Bz-Aw} \mathcal{J}_0(z, w) \\ &+ e^{-Aw} \frac{e^{-Bz} - e^{L_p z}}{z} g(w) + e^{-Bz} \frac{e^{-Aw} - e^{L_q w}}{w} g(z) \\ &+ \frac{1 - e^{-Bz-Aw} - e^{L_p z} + e^{L_p z - Aw} - e^{L_q w} + e^{L_q w - Bz}}{zw}. \end{aligned} \quad (3.28)$$

Every denominator on the right is removable; the last numerator vanishes on both coordinate axes. Equivalently, the full separated-grid coefficient formula of the incoming report remains valid at collision after replacing $I_{ij}(\theta)$ by $J_{ij;0,0}^{1,1}$ and both single Stieltjes arguments $\theta, 1 - \theta$ by 1. This substitution is a conclusion of local subtraction, not evaluation of a divergent integrand at a singular shift.

For completeness, the scalar correction behind (3.28) is visible before extracting coefficients:

$$\mathcal{J}_{0,0}^{p,q} = \mathcal{C}_{0,0}^{p,q} + \frac{1}{zw} - \frac{e^{L_p z}}{z} \left(e^{-Aw} \zeta(1-w) + \frac{1}{w} \right) - \frac{e^{L_q w}}{w} \left(e^{-Bz} \zeta(1-z) + \frac{1}{z} \right).$$

The terms in parentheses are holomorphic by cancellation of the zeta pole. The additional $1/(zw)$ comes from the two scalar terms used to define $\mathcal{G}$. Expanding this identity and its undilated version proves (3.28).

In particular,

$$\text{FP}_x \int_0^1 \psi(\{px\}) \psi(\{qx\}) dx = 2\gamma_1 - 2\zeta(2) - 2\gamma K - K^2 + \log p \log q. \quad (3.29)$$

Indeed, (3.14) gives $T(z, w) = 1 + 2\zeta(2)zw + O((|z| + |w|)^3)$, so (3.27) yields $J_{00;0,0}^{1,1} = 2\gamma_1 - 2\zeta(2)$. This recovers the undilated coincident identity in the incoming report. The occurrence of K in the colliding unequal case is forced by the ambient subtraction scale.

3.5 Exact matching with the noncollision expansion

Let

$$F_{mn;r,k}^{p,q}(\theta) = \text{FP}_x \int_0^1 \gamma_m^{(r)}(\{px\}) \gamma_n^{(k)}(\{qx + a\}) dx, \quad \theta = \{Pa\} \in (0, 1),$$

denote the separated finite part. Its dependence on a factors through θ by the finite covering formula, or by Fourier orthogonality. The next theorem resolves the collision matching question, including every mixed derivative and Stieltjes index. Set

$$C_N(z, w) = \frac{\Gamma(N+1-z-w)\Gamma(1+z)}{\Gamma(1-w)}.$$

Unlike V_N , this definition is also regular for $N = 0$.

Theorem 3.5 (Convergent collision expansion). *For $m, n, r, k \geq 0$, define the explicit logarithmic polynomial*

$$\mathcal{P}_{mn;r,k}^{p,q}(L) = (-1)^k Q^r P^k m! n! [z^m w^n] \frac{e^{-Bz-Aw+wL} C_N(z, w) e^{zL} - e^{Kz} p_r(z) (1-w)_N}{z}. \quad (3.30)$$

The quotient is holomorphic. The polynomial has degree $m + n + 1$ and leading coefficient

$$\frac{(-1)^k N! Q^r P^k}{m+1}.$$

There is an exact expansion

$$F_{mn;r,k}^{p,q}(\theta) = \frac{\mathcal{P}_{mn;r,k}^{p,q}(\log \theta)}{\theta^{N+1}} + J_{mn;r,k}^{p,q} + \sum_{j \geq 1} h_j \theta^j, \quad 0 < \theta < 1, \quad (3.31)$$

whose regular power series converges locally uniformly for $|\theta| < 1$. Thus the regular constant at collision agrees exactly with the independently defined merged-grid finite part.

Proof. The same Fourier and local-subtraction proof gives the separated kernel by replacing $V_N(z, w)$ with $C_N(z, w)\zeta(N+1-z-w, \theta)$ and $W_N(w)$ with $(1-w)_N\zeta(N+1-w, \theta)$ in the first divided difference, and by making the analogous replacements with $1-\theta$ in the second. When $N=0$, the additional pure scalar terms containing $1/(zw)$ are independent of θ and remain unchanged; the Hurwitz terms inside the contact numerators take the same θ or $1-\theta$ replacements. These statements follow directly from the scalar formula preceding (3.29).

Use the exact shift and Taylor identities

$$\begin{aligned}\zeta(s, \theta) &= \theta^{-s} + \zeta(s, 1 + \theta), \\ \zeta(s, 1 + \theta) &= \sum_{j \geq 0} \frac{(-1)^j (s)_j}{j!} \zeta(s + j) \theta^j, \\ \zeta(s, 1 - \theta) &= \sum_{j \geq 0} \frac{(s)_j}{j!} \zeta(s + j) \theta^j.\end{aligned}$$

The first isolated power produces exactly (3.30). The $j=0$ regular terms give the colliding kernel proved above. For $j \geq 1$ there is no zeta pole near the spectral origin, and the coefficient-generating function is

$$\begin{aligned}\mathcal{H}_j(z, w) &= \frac{Q^r P^k e^{-Bz-Aw}}{j!} \left\{ (-1)^{k+j} \frac{V_{N+j}(z, w) - e^{Kz} p_r(z) W_{N+j}(w)}{z} \right. \\ &\quad \left. + (-1)^r \frac{V_{N+j}(w, z) - e^{Kw} p_k(w) W_{N+j}(z)}{w} \right\}.\end{aligned}\quad (3.32)$$

Here $h_j = m!n![z^m w^n] \mathcal{H}_j(z, w)$. After division by $j!$, the gamma factors and rising factorials grow at most polynomially in j on a fixed sufficiently small spectral polydisc; their spectral derivatives introduce powers of $\log j$. Thus $|\theta|^j$ gives locally uniform convergence for $|\theta| < 1$, before and after any fixed coefficient extraction. For $N=0$, the common $j=0$ spectral pole is removed in the complete kernel (3.28); it does not affect the convergent $j \geq 1$ series.

Finally the highest L -degree in (3.30) comes from $C_N(0, 0) = N!$ and $e^{(z+w)L}/z$. The second term in that numerator contains L only through e^{wL} and cannot contribute degree $m+n+1$. Coefficient extraction gives the asserted leading coefficient. $\square$

The lowest polynomial is already informative:

$$\mathcal{P}_{00;r,k}^{p,q}(L) = (-1)^k N! Q^r P^k (L - H_N + H_r - K). \quad (3.33)$$

For example,

$$\lim_{\theta \downarrow 0} \left\{ \text{FP}_x \int_0^1 \psi(\{2x\}) \psi'(\{3x+a\}) dx + \frac{2(\log \theta - 1 - \log 6)}{\theta^2} \right\} = \frac{\pi^2}{3}, \quad \theta = \{2a\}.$$

The finite term is therefore fixed by an exact convergent expansion, including the ambient scale and every local subtraction term.

3.6 Common dilation and an integration safeguard

For $N \geq 1$ and a further positive integer e , put $D = \log e$. The reduced pair P, Q is unchanged and K becomes $K + D$. From the master kernel,

$$\begin{aligned}\mathcal{J}_{r,k}^{ep,eq} - \mathcal{J}_{r,k}^{p,q} &= -Q^r P^k e^{-Bz-Aw} \left\{ (-1)^k p_r(z) W_N(w) e^{Kz} \frac{e^{Dz} - 1}{z} \right. \\ &\quad \left. + (-1)^r p_k(w) W_N(z) e^{Kw} \frac{e^{Dw} - 1}{w} \right\}.\end{aligned}\quad (3.34)$$

This is a finite polynomial in D at every fixed pair of Stieltjes indices. At zero index and odd N , the two terms cancel, explaining the exact scale independence in (3.19). At even N the change is the negative of D times the summed local t^{-1} coefficient.

The incoming separated-grid theorem is valid as stated and requires no retraction. Its collision question can now be marked resolved by (3.10), (3.28), and (3.31). What must not be done during integration is to multiply the two grid-supported contact distributions at collision, or to keep only residues in the local spectral corrections. The pointwise product has the unambiguous ambient finite part proved here; its analytic representation requires the complete pole numerators.

4 Nonlinear endpoint coordinates: a finite exact law

The two recent dilation reports explicitly ask for the replacement of their affine coordinate pt by $pt + q_2t^2 + q_3t^3 + \dots$ [3, 6]. This section answers that question for every Stieltjes index and every argument derivative. The answer is local: a singularity at any other point is treated after translation. All finite parts use a right-hand cutoff in the indicated coordinate. Two-sided symmetric prescriptions are different objects.

Let f be a real analytic local diffeomorphism with

$$f(0) = 0, \quad f'(0) = p > 0, \quad H_f(t) = \frac{f(t)}{t} = p + q_2t + q_3t^2 + \dots. \quad (4.1)$$

Choose a sufficiently small neighborhood so that H_f is positive on the real axis. The power $H_f(t)^z$ uses the analytic logarithm whose value at zero is $\log p$. For a distribution T the scalar pullback is

$$\langle f^*T, \varphi \rangle = \left\langle T, \frac{\varphi(f^{-1}(y))}{f'(f^{-1}(y))} \right\rangle. \quad (4.2)$$

Thus $f^*T = T \circ f$ for ordinary functions. The Jacobian in this definition is essential.

4.1 A general coefficient lemma

Write $\delta^{(j)}$ for the j th distributional derivative at zero, so that $\langle \delta^{(j)}, \varphi \rangle = (-1)^j \varphi^{(j)}(0)$. For integers $q \geq 1$ and $n \geq 0$ set

$$U_{q,n} = \text{FP}_y[\mathbf{1}_{y>0} y^{-q} \log^n y].$$

Only the germ at zero matters; a smooth cutoff may be inserted away from zero in every formula below.

Theorem 4.1 (Coordinate defect for a logarithmic pole). *One has*

$$f^*U_{q,n} = \text{FP}_t[\mathbf{1}_{t>0} f(t)^{-q} \log^n f(t)] + \mathcal{C}_f(q, n), \quad (4.3)$$

where the complete correction is the finite distribution

$$\mathcal{C}_f(q, n) = n! \sum_{j=0}^{q-1} \frac{(-1)^j}{j!} [z^{n+1} t^{q-1-j}] H_f(t)^{z-q} \delta^{(j)}. \quad (4.4)$$

It depends only on $f'(0), \dots, f^{(q)}(0)$.

Proof. Begin with the locally integrable meromorphic family y_+^{z-q} for $\Re z > q-1$. Pairing its pullback with a test function gives

$$\int_0^h t^{z-q} H_f(t)^{z-q} \varphi(t) dt.$$

Subtract the Taylor polynomial in t through degree $q - 1$ from the holomorphic factor $H_f(t)^{z-q}\varphi(t)$. The remainder is locally normally integrable for $\Re z > -1$. A Taylor term of degree a has integral

$$g_a(z) \frac{h^{z-q+a+1}}{z - q + a + 1}.$$

Near $z = 0$, only $a = q - 1$ has a pole. The other Taylor terms have nonzero denominators at zero, and taking their spectral coefficients commutes with the coordinate finite part. For the exceptional term, the cutoff expression is

$$g_{q-1}(z) \frac{h^z - \varepsilon^z}{z}.$$

The constant term in ε of its coefficient of z^n is the regular coefficient of $g_{q-1}(z)h^z/z$, less $[z^{n+1}]g_{q-1}(z)$. In the original y coordinate that last correction is zero, because its Taylor coefficient is independent of z . Therefore the difference after pullback is

$$n![z^{n+1}t^{q-1}]H_f(t)^{z-q}\varphi(t).$$

Expanding the test function and using the convention for $\delta^{(j)}$ proves (4.4). Its largest required power of t is $q - 1$, proving the finite dependence assertion. $\square$

Every function with a finite expansion in negative integer powers and nonnegative logarithmic powers is covered by linearity. Terms $t^a \log^b t$ with $a \geq 0$ are locally integrable and create no defect. Consequently the theorem also applies to products of singular germs after their ordinary local expansion has been performed; it does not require a multiplication of distributions.

Corollary 4.2 (A sharp disappearance threshold). *If $f'(0) = 1$, then $\mathcal{C}_f(q, n) = 0$ for $n \geq q - 1$. For $q \geq 2$, writing $f(t) = t + at^2 + O(t^3)$, one has*

$$\mathcal{C}_f(q, q - 2) = \frac{a^{q-1}}{q - 1} \delta. \quad (4.5)$$

Thus the threshold is attained when $a \neq 0$.

Proof. When $H_f(0) = 1$, $[t^k]H_f(t)^{z-q}$ is a polynomial in z of degree at most k . This follows by expanding $\exp((z - q) \log H_f(t))$, since $\log H_f(t) = O(t)$. In (4.4), $k \leq q - 1$, so its coefficient of z^{n+1} vanishes for $n \geq q - 1$. At $n = q - 2$, only $j = 0$ can survive; the leading coefficient of $[t^{q-1}]H_f(t)^{z-q}$ is $a^{q-1}/(q - 1)!$. Multiplication by $(q - 2)!$ proves the formula. $\square$

4.2 Composition is an exact cocycle

For any logarithmic singular germ u define

$$C_f(u) = f^*(\text{FP } u) - \text{FP}(u \circ f).$$

Theorem 4.1 computes this distribution from the finite singular expansion of u . The expansion of $u \circ f$ is again of the same kind, up to locally integrable terms.

Theorem 4.3 (Composition and explicit delta pullback). *For orientation-preserving analytic local coordinates f, g ,*

$$C_{f \circ g}(u) = g^*C_f(u) + C_g(u \circ f). \quad (4.6)$$

The delta derivatives in this formula can be computed without series reversion:

$$f^*\delta^{(r)} = \sum_{j=0}^r \frac{(-1)^{r-j} r!}{j!} [t^{r-j}]H_f(t)^{-r-1} \delta^{(j)}. \quad (4.7)$$

Proof. Pullbacks satisfy $(f \circ g)^* = g^* f^*$. Insert and subtract $g^* \text{FP}(u \circ f)$ in the definition of $C_{f \circ g}$; this gives (4.6) exactly. For (4.7), take the residue at $z = 0$ in the proof of Theorem 4.1 with $q = r + 1$. The residue of y_+^{z-r-1} is $(-1)^r \delta^{(r)}/r!$. The residue after pullback pairs with φ as $[t^r]H_f(t)^{-r-1}\varphi(t)$. Expanding φ gives the displayed formula. $\square$

This proves the composition requirement in the source question, rather than merely checking it for low-degree maps. The accompanying exact code also compares both sides using independently composed rational Taylor series.

4.3 Every Stieltjes index and every argument derivative

Let G_n be the canonical periodic finite part of γ_n in the unit coordinate. Its local meromorphic generator is

$$Z(1-z) = \frac{\delta_0 - 1}{z} + \sum_{n \geq 0} G_n \frac{z^n}{n!}. \quad (4.8)$$

Indeed, locally across the endpoint, $\zeta(1-z, \{y\}) = y_+^{z-1} + \zeta(1-z, 1+y)$. The second term is smooth in y and has only the constant spectral pole $-1/z$. Set

$$E_r(z) = \prod_{h=1}^r (1 - z/h), \quad E_0(z) = 1. \quad (4.9)$$

Theorem 4.4 (Nonlinear Stieltjes contact law). *For all $n, r \geq 0$, as local distributions,*

$$f^*(D_y^r G_n) = \text{FP}_t[\gamma_n^{(r)}(\{f(t)\})] + A_{n,r}(f), \quad (4.10)$$

where

$$A_{n,r}(f) = n!r! \sum_{j=0}^r \frac{(-1)^{r+j}}{j!} [z^{n+1}t^{r-j}] E_r(z) H_f(t)^{z-r-1} \delta^{(j)}. \quad (4.11)$$

The correction depends only on the $(r+1)$ -jet of f .

Proof. The singular part of $D_y^r Z(1-z)$ is $(-1)^r (1-z)_r y_+^{z-r-1}$. After pullback, pair it with a test function and repeat the Taylor subtraction proof above. Only its degree- r Taylor coefficient creates a pole at $z = 0$; the correction is

$$n![z^{n+1}t^r](-1)^r (1-z)_r H_f(t)^{z-r-1} \varphi(t).$$

Since $(1-z)_r = r!E_r(z)$, this is (4.11). The smooth local Hurwitz term commutes with coefficient extraction and contributes no contact distribution. The largest Taylor order needed is r , which proves the jet assertion. $\square$

For an affine germ $f(t) = pt$, only the term $j = r$ remains, and

$$A_{n,r}(pt) = p^{-r-1} b_{n,r}(\log p) \delta^{(r)}, \quad b_{n,r}(L) = n![z^{n+1}] e^{Lz} E_r(z). \quad (4.12)$$

After multiplication by p^r , this is precisely the earlier ambient-derivative dilation law. In particular,

$$b_{0,r}(L) = L - H_r, \quad b_{1,r}(L) = \frac{L^2}{2} - H_r L + \frac{H_r^2 - H_r^{(2)}}{2}.$$

The coefficient algebra is therefore an explicit extension of the source's harmonic contact polynomial, with the same normalization.

For $f(t) = pt + q_2 t^2 + O(t^3)$ and $L = \log p$, the first two argument orders take the particularly small form

$$A_{n,0}(f) = \frac{L^{n+1}}{(n+1)p} \delta, \quad (4.13)$$

$$A_{n,1}(f) = \frac{n!}{p^2} [z^{n+1}] (1-z) e^{Lz} \delta' + \frac{n!q_2}{p^3} [z^{n+1}] (2-3z+z^2) e^{Lz} \delta. \quad (4.14)$$

Curvature creates an actual lower-order contact term; replacing f by its tangent line loses it.

Corollary 4.5 (Sharp unit-tangent Stieltjes threshold). *Suppose $f(t) = t + at^2 + O(t^3)$ and $r \geq 1$. Then*

$$A_{n,r}(f) = 0 \quad (n \geq 2r), \quad A_{2r-1,r}(f) = \frac{(2r-1)!}{r!} a^r \delta. \quad (4.15)$$

Proof. The degree in z of $[t^{r-j}]E_r(z)H_f(t)^{z-r-1}$ is at most $2r-j$. This proves the vanishing assertion. At $n = 2r-1$ only $j = 0$ survives. The leading coefficients are $(-1)^r/r!$ for E_r and $a^r/r!$ for $[t^r]H_f^{z-r-1}$. Substitute them into (4.11) to obtain the second formula. $\square$

For $f(t) = t + at^2 + bt^3 + O(t^4)$ some exact coefficients are

$$\begin{aligned} A_{0,1} &= -\delta' - 3a\delta, & A_{1,1} &= a\delta, \\ A_{0,2} &= -\frac{3}{2}\delta'' - 11a\delta' + (11b - 25a^2)\delta, & A_{3,2} &= 3a^2\delta. \end{aligned} \quad (4.16)$$

Together with (4.15), these examples show why the earlier zero contact terms at unit affine scale cannot simply be transported through a nonlinear coordinate.

4.4 Exact antiderivatives and their endpoint constants

There is a useful scalar consequence which does not require any distribution notation. Suppose f is increasing on $[0, b]$, $f(0) = 0$, $f'(0) > 0$, and $B = f(b) > 0$. Assume that the Taylor germ at zero is as above. For $r \geq 1$ define

$$\mathcal{E}_{n,r}(f) = \gamma_n^{(r-1)}(1) + (-1)^{r-1} n! [z^n t^r] (1-z)_{r-1} H_f(t)^{z-r}. \quad (4.17)$$

Theorem 4.6 (Finite-part primitive endpoint identity). *For every $n \geq 0$ and $r \geq 1$,*

$$\text{FP}_t \int_0^b f'(t) \gamma_n^{(r)}(f(t)) dt = \gamma_n^{(r-1)}(B) - \mathcal{E}_{n,r}(f). \quad (4.18)$$

For argument order zero the exact change-of-coordinate identity is

$$\text{FP}_t \int_0^b f'(t) \gamma_n(f(t)) dt = \text{FP}_y \int_0^B \gamma_n(y) dy - \frac{\log^{n+1} p}{n+1}. \quad (4.19)$$

Proof. At a positive cutoff, the fundamental theorem of calculus gives $\gamma_n^{(r-1)}(B) - \gamma_n^{(r-1)}(f(\varepsilon))$. Use

$$\gamma_n(y) = y^{-1} \log^n y + \gamma_n(1+y)$$

and differentiate $r-1$ times. The smooth term has constant $\gamma_n^{(r-1)}(1)$. Write its singular term as the coefficient of $z^n/n!$ in $(-1)^{r-1}(1-z)_{r-1}t^{z-r}H_f(t)^{z-r}$. The constant term in t comes only from the Taylor term of degree r , with no power of $\log t$ retained. This is exactly (4.17). For $r = 0$, substitution gives a lower endpoint primitive $\log^{n+1} f(\varepsilon)/(n+1)$, whose constant term is $\log^{n+1} p/(n+1)$. All smooth endpoint contributions agree in the two coordinates, proving (4.19). $\square$

For example, put $f(t) = t + ct^2$ on an interval where it is increasing. Then $B = b + cb^2$ and the formulas give

$$\text{FP} \int_0^b (1 + 2ct) \psi'(t + ct^2) dt = \psi(B) + \gamma - c, \quad (4.20)$$

$$\text{FP} \int_0^b (1 + 2ct) \psi''(t + ct^2) dt = \psi'(B) - \zeta(2) - 3c^2, \quad (4.21)$$

$$\text{FP} \int_0^b (1 + 2ct) \gamma_1'(t + ct^2) dt = \gamma_1(B) - \gamma_1(1) - c. \quad (4.22)$$

These are exact antiderivative identities with explicit curvature constants. In particular, a formal substitution which silently equates the t and $f(t)$ finite parts gives an incorrect constant.

The zero-order Hurwitz primitive on the right of (4.19) can be written, if desired, by integrating $\zeta(1-z, y)$ as $\zeta(-z, y)/z$ and taking its regular spectral coefficients. At $n = 0$, $\gamma_0 = -\psi$ gives the familiar logarithmic Gamma primitive. The new ingredient above is the exact endpoint transformation, not the classical Hurwitz primitive itself [8].

5 Three unequal frequencies and all their regularized jets

The current incoming report treats three distinct translates and asks for three unequal integer frequencies [6]. Fourier orthogonality leaves a lattice of rank two in this problem. Finite congruence filters nevertheless give an explicit finite family of colored Tornheim kernels. Local subtraction then converts the spectral identity into an all-index Stieltjes identity in the ambient coordinate. The classical use of colored Tornheim sums and their polylogarithmic reductions is credited to the literature [9]; the formulas here address the unequal-frequency and fixed-coordinate question from the supplied research.

Let p_1, p_2, p_3 be positive integers and a_1, a_2, a_3 be real shifts, considered modulo one. Write

$$S_i = \{x \in \mathbb{R}/\mathbb{Z} : p_i x + a_i \in \mathbb{Z}\}, \quad e(t) = e^{2\pi i t}.$$

Throughout this section assume that S_1, S_2, S_3 are pairwise disjoint. Equivalently, for every $i \neq j$,

$$p_j a_i - p_i a_j \notin \gcd(p_i, p_j)\mathbb{Z}. \quad (5.1)$$

At each grid point the cutoff is the right ambient coordinate $t = x - x_0$; the cutoffs at different grid points can tend to zero independently.

5.1 Weighted colored kernels and finite character filters

For positive integers a, b and nontrivial unit complex numbers X, Y define initially in an absolutely convergent half-space

$$T_{a,b}(A, B, C; X, Y) = \sum_{m,n \geq 1} \frac{X^m Y^n}{m^A n^B (am + bn)^C}. \quad (5.2)$$

The positive real denominators use real logarithms. For $\Re C > 0$ the entire continuation in A, B is given by

$$T_{a,b}(A, B, C; X, Y) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \text{Li}_A(Xe^{-at}) \text{Li}_B(Ye^{-bt}) dt. \quad (5.3)$$

In fact this is jointly entire in all three orders after continuation. To prove the statement, subtract any finite Taylor polynomial at $t = 0$ from the product of polylogarithms. It is holomorphic there, locally uniformly in A, B , because $X, Y \neq 1$. The Taylor integrals have poles only at $C = 0, -1, \dots$, each canceled by $1/\Gamma(C)$. The integral over $[1, \infty)$ converges locally normally for all orders because its integrand decays exponentially. Increasing the Taylor order proves entire continuation and licenses every fixed mixed order derivative.

These weighted kernels introduce no extra class of functions beyond ordinary colored Tornheim kernels. Choose any roots $\xi^a = X$ and $\eta^b = Y$. Applying two finite root filters gives

$$T_{a,b}(A, B, C; X, Y) = a^{A-1} b^{B-1} \sum_{r=0}^{a-1} \sum_{s=0}^{b-1} T_{1,1}(A, B, C; \xi e(r/a), \eta e(s/b)). \quad (5.4)$$

Different choices of roots merely permute the summands. None of these twists is one, since $X, Y \neq 1$. The proof first changes variables $u = am, v = bn$ in the absolutely convergent series, and then uses $a^{-1} \sum_r e(ru/a) = \mathbf{1}_{a|u}$ and its b analogue. Analytic continuation proves the identity at all orders.

For each $k \in \{1, 2, 3\}$ let $i < j$ be the remaining indices, and for $h = 0, \dots, p_k - 1$ put

$$\begin{aligned} X_{ik,h} &= e\left(a_i - \frac{p_i a_k}{p_k} + \frac{h p_i}{p_k}\right), \\ X_{jk,h} &= e\left(a_j - \frac{p_j a_k}{p_k} + \frac{h p_j}{p_k}\right), \\ \vartheta_k &= \frac{\pi}{2}(\lambda_i + \lambda_j - \lambda_k). \end{aligned} \quad (5.5)$$

Every twist in (5.5) is nontrivial. Indeed, $X_{ik,h} = 1$ for some integer h is equivalent to $p_k a_i - p_i a_k \in \gcd(p_i, p_k)\mathbb{Z}$, which contradicts (5.1).

Theorem 5.1 (Finite spectral formula for three unequal frequencies). *For $\Re \lambda_i > 0$,*

$$K(\boldsymbol{\lambda}; \mathbf{p}, \mathbf{a}) = \int_0^1 \prod_{i=1}^3 \zeta(1 - \lambda_i, \{p_i x + a_i\}) dx = \frac{\prod_i \Gamma(\lambda_i)}{(2\pi)^{\sum_i \lambda_i}} \mathcal{S}(\boldsymbol{\lambda}), \quad (5.6)$$

where the finite sum is

$$\begin{aligned} \mathcal{S}(\boldsymbol{\lambda}) = \sum_{k=1}^3 p_k^{\lambda_k-1} \sum_{h=0}^{p_k-1} \{ & e^{-i\vartheta_k} T_{p_i, p_j}(\lambda_i, \lambda_j, \lambda_k; X_{ik,h}, X_{jk,h}) \\ & + e^{i\vartheta_k} T_{p_i, p_j}(\lambda_i, \lambda_j, \lambda_k; X_{ik,h}^{-1}, X_{jk,h}^{-1}) \}. \end{aligned} \quad (5.7)$$

The right side is a meromorphic continuation to all three order variables. It has $2(p_1 + p_2 + p_3)$ weighted-kernel terms before symmetry or character simplifications.

Proof. Start with $\Re \lambda_i > 1$, where each Hurwitz Fourier series is absolutely convergent. For a nonzero frequency n , its coefficient is

$$\frac{\Gamma(\lambda_i)}{(2\pi|n|)^{\lambda_i}} e^{-i\pi \lambda_i \operatorname{sgn}(n)/2}.$$

After dilation and translation, integration retains precisely $p_1 n_1 + p_2 n_2 + p_3 n_3 = 0$, with all $n_i \neq 0$. There are six disjoint sign cones. In the cone with $n_i = m > 0$, $n_j = n > 0$ and $n_k < 0$, one has

$$-n_k = \frac{p_i m + p_j n}{p_k} \in \mathbb{N}.$$

The denominator contributes $p_k^{\lambda_k}$ and the phase of the Fourier coefficient is $e^{-i\vartheta_k}$. Insert

$$\mathbf{1}_{p_k | p_i m + p_j n} = \frac{1}{p_k} \sum_{h=0}^{p_k-1} e\left(\frac{h(p_i m + p_j n)}{p_k}\right).$$

Combining this filter with the translation phases produces exactly the first term in braces in (5.7), including the factor $p_k^{\lambda_k-1}$. Reversing all frequency signs gives the second term, after reindexing the finite filter by $h \mapsto -h$. Each lattice point occurs once.

The original integral is jointly holomorphic for $\Re \lambda_i > 0$: near a grid point only one factor is singular, with a locally integrable majorant $t^{\sigma-1}$ on any compact parameter set. Order derivatives add only logarithmic powers. Entire continuation of the colored kernels makes the right side holomorphic in this half-space as well. The identity theorem proves the asserted extension, and the Gamma factors give the meromorphic continuation. $\square$

There is also a longer formula directly in the known equal-frequency kernel. The ordinary Hurwitz multiplication law gives

$$K(\boldsymbol{\lambda}; \mathbf{p}, \mathbf{a}) = \prod_i p_i^{\lambda_i-1} \sum_{h_1=0}^{p_1-1} \sum_{h_2=0}^{p_2-1} \sum_{h_3=0}^{p_3-1} K\left(\boldsymbol{\lambda}; (1, 1, 1), \left(\frac{a_i + h_i}{p_i}\right)_{i=1}^3\right). \quad (5.8)$$

All three translated endpoints in every summand are distinct. This provides an alternative finite reduction and confirms that the rank-two lattice does not require a new, unbounded family of kernels.

5.2 All Stieltjes and argument jets in the ambient coordinate

Set $\mathcal{G}(\lambda, y) = \zeta(1 - \lambda, y) + 1/\lambda$ as before, and define the pair kernels by

$$C_{ij}(\lambda_i, \lambda_j) = \int_0^1 \zeta(1 - \lambda_i, \{p_i x + a_i\}) \zeta(1 - \lambda_j, \{p_j x + a_j\}) dx,$$

with their ordinary separated-grid analytic continuation. They have the finite Gamma–polylogarithm expression (2.10). All shift derivatives below are taken within a local chamber of separated real shifts. Define

$$\mathcal{L}(\boldsymbol{\lambda}) = K(\boldsymbol{\lambda}) + \sum_{\substack{\{i,j,k\}=\{1,2,3\} \\ i < j}} \frac{C_{ij}(\lambda_i, \lambda_j)}{\lambda_k} + \frac{1}{\lambda_1 \lambda_2 \lambda_3}. \quad (5.9)$$

In an ordinary convergent domain this is the integral of the product of the three $\mathcal{G}$ functions. The terms with just one Hurwitz factor vanish because its integral over a period is zero.

For $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{N}_0^3$ let $\partial_{\mathbf{a}}^{\mathbf{r}} = \prod_i \partial_{a_i}^{r_i}$. For $x_0 \in S_i$ and $j \neq i$ put $\eta_{ij}(x_0) = \{p_j x_0 + a_j\} \in (0, 1)$. Define the finite local numerator

$$B_i^{\mathbf{r}}(\boldsymbol{\lambda}) = (-1)^{r_i} (1 - \lambda_i)_{r_i} p_i^{\lambda_i - r_i - 1} \cdot \sum_{x_0 \in S_i} [t^{r_i}] \prod_{j \neq i} \partial_y^{r_j} \mathcal{G}(\lambda_j, \eta_{ij}(x_0) + p_j t). \quad (5.10)$$

It is holomorphic near the spectral origin. It uses a finite Taylor polynomial and values of Stieltjes functions and their argument derivatives at nonsingular points.

Theorem 5.2 (All-index triple finite-part identity). *The function*

$$\mathcal{J}_{\mathbf{r}}(\boldsymbol{\lambda}) = \partial_{\mathbf{a}}^{\mathbf{r}} \mathcal{L}(\boldsymbol{\lambda}) - \sum_{i=1}^3 \frac{B_i^{\mathbf{r}}(\boldsymbol{\lambda})}{\lambda_i} \quad (5.11)$$

is holomorphic in a sufficiently small neighborhood of the origin. For all $m_1, m_2, m_3 \geq 0$,

$$\begin{aligned} \text{FP}_x \int_0^1 \prod_{i=1}^3 \gamma_{m_i}^{(r_i)}(\{p_i x + a_i\}) dx \\ = \left(\prod_i m_i! \right) [\lambda_1^{m_1} \lambda_2^{m_2} \lambda_3^{m_3}] \mathcal{J}_{\mathbf{r}}(\boldsymbol{\lambda}). \end{aligned} \quad (5.12)$$

Together with (5.7), this is an explicit finite colored-Tornheim, Gamma, and local Stieltjes coefficient formula for every choice of the six indices.

Proof. Begin with $\Re \lambda_i > r_i + 1$. Argument differentiation and integration then commute, including at moving grid points: the singular pieces vanish to the required orders and the regular pieces match across each periodic endpoint, so the boundary terms cancel. Hence $\partial_{\mathbf{a}}^{\mathbf{r}} \mathcal{L}$ is the meromorphic integral of the product of the three argument-derivative generators.

Near $x_0 \in S_i$ the singular term of factor i is

$$(-1)^{r_i} (1 - \lambda_i)_{r_i} p_i^{\lambda_i - r_i - 1} t^{\lambda_i - r_i - 1}.$$

The other two factors are smooth there. Subtract their Taylor polynomial through degree r_i . The degree- r_i term alone has denominator λ_i near the origin; summed over S_i its numerator is exactly (5.10). As in Theorem 4.1, its meromorphic integral over $(0, h)$ is $B_i h^{\lambda_i} / \lambda_i$, whereas the generating function of its coordinate finite parts is $B_i (h^{\lambda_i} - 1) / \lambda_i$. Their difference is the complete quantity B_i / λ_i .

The lower Taylor degrees have denominators bounded away from zero on a small spectral polydisc. The remainders are locally normally integrable there. A partition of unity treats the finitely many disjoint singular points independently. Subtracting all three complete numerators therefore leaves a holomorphic function whose Taylor coefficients are exactly the stated finite-part integrals. The factorials in (5.12) follow from the exponential Stieltjes generating convention. $\square$

For zero argument derivatives the local term simplifies to

$$B_i^0(\boldsymbol{\lambda}) = p_i^{\lambda_i-1} \sum_{x_0 \in S_i} \prod_{j \neq i} \mathcal{G}(\lambda_j, \eta_{ij}(x_0)).$$

Even here its full dependence on λ_i must be retained; using only $B_i(0, \lambda_j, \lambda_k)$ omits the powers of $\log p_i$. Thus the character calculation and the ambient-coordinate theorem are separate necessary parts of the reduction.

5.3 Independent polynomial checks

At positive integer λ_i , the identity $\zeta(1 - \lambda_i, y) = -B_{\lambda_i}(y)/\lambda_i$ makes (5.6) a rational piecewise-polynomial integral whenever the shifts are rational. This provides a check independent of Fourier or special-function quadrature. For

$$(p_1, p_2, p_3) = (1, 2, 3), \quad (a_1, a_2, a_3) = (0, 1/5, 1/7),$$

the exact integrations give

$$\begin{aligned} K(2, 2, 2) &= -\frac{49960598339}{1715322420000000}, \\ K(2, 3, 2) &= -\frac{1641301145317}{315190494675000000}. \end{aligned} \quad (5.13)$$

The supplied program compares these exact rational values with the independent filtered Mellin integrals (5.3). The spectral check does not replace the local-subtraction proof for Stieltjes jets, which is given above.

6 Fully shifted height-one sums: an entire difference and exact Laurent data

The inner and outer shifts of a nested harmonic sum are different pieces of data. The earlier shifted height-one families in this project move the outer denominator while retaining ordinary inner harmonic numbers. The newest nested-jet report instead gives all nested denominators a common variable exponent. Here every denominator has the same shift, while only the outer exponent varies. A Gauss connection formula shows that changing the common shift preserves a universal singular part, up to an explicit Gamma factor. The remaining function is entire in the spectral variable, and its values at every nonpositive integer are finite Stirling transforms.

At depth one this family belongs to the harmonic Hurwitz-zeta literature of Kargin, Dil, Cenkci and Can [10]; the recent preprint of Kara Öztürk and Can treats the two-parameter depth-one family and negative Laurent coefficients [11]. The latter comparison is based on its retrieved abstract: its complete text was not available in this research environment. Thus the depth-one meromorphic continuation and negative-integer evaluation problem are not claimed as new. The all-depth entire difference, finite Stirling rule, and complete pole-order classification below were derived here and were not found in the inspected repository material; priority over all literature is not asserted.

6.1 The common-shift convention

Let $\Re a > 0$. All powers of $n + a$ use the principal logarithm, and u will remain in a sufficiently small disc centered at zero. Set

$$e_r(n; a) = [u^r] \prod_{j=0}^{n-1} \left(1 + \frac{u}{j+a}\right), \quad e_0(n; a) = 1, \quad (6.1)$$

$$M_r(s; a) = \sum_{n=0}^{\infty} \frac{e_r(n; a)}{(n+a)^s}, \quad \Re s > 1, \quad (6.2)$$

$$F_a(s, u) = \sum_{n=0}^{\infty} \frac{(a+u)_n}{(a)_n(n+a)^s} = \sum_{r \geq 0} M_r(s; a) u^r. \quad (6.3)$$

The first product is empty when $n = 0$. Equivalently,

$$M_r(s; a) = \sum_{n_0 > n_1 > \dots > n_r \geq 0} \frac{1}{(n_0 + a)^s (n_1 + a) \cdots (n_r + a)}.$$

Thus $M_0(s; a) = \zeta(s, a)$ and $M_r(s; 1) = \zeta(s, \{1\}^r)$. If $H_n^{(k)}(a) = \sum_{j=0}^{n-1} (j+a)^{-k}$, then

$$e_1(n; a) = H_n^{(1)}(a), \quad e_2(n; a) = \frac{(H_n^{(1)}(a))^2 - H_n^{(2)}(a)}{2}.$$

The general e_r is the corresponding elementary-symmetric Bell polynomial. Uniformly on compact a -sets, $e_r(n; a) = O((1 + \log n)^r)$; hence every fixed-depth series converges normally for $\Re s > 1$. The Gamma-ratio Dirichlet series in (6.3) converges normally when $\Re s > 1 + \Re u$, locally uniformly in its parameters. Interchange with its depth Taylor series is justified, for example, on $\Re s > 1 + |u|$; at each fixed $\Re s > 1$ one may choose a sufficiently small disc in u . The direct-series estimate follows from

$$\frac{(a+u)_n}{(a)_n} = \frac{\Gamma(a)}{\Gamma(a+u)} \frac{\Gamma(n+a+u)}{\Gamma(n+a)} = O(n^{\Re u}).$$

Gamma-ratio asymptotics and their uniform sector versions are classical [14].

Introduce

$$C_a(u) = \frac{\Gamma(a)}{\Gamma(a+u)} = \sum_{j \geq 0} c_j(a) u^j, \quad (6.4)$$

$$K_a(u) = \Gamma(1+u) C_a(u) = \sum_{j \geq 0} k_j(a) u^j. \quad (6.5)$$

The Gamma logarithm is analytic on the right half-plane, so

$$C_a(u) = \exp \left(- \sum_{j \geq 1} \frac{\psi^{(j-1)}(a)}{j!} u^j \right). \quad (6.6)$$

Consequently every coefficient is a finite polynomial in polygamma functions. In particular, with $P = \psi(a)$, $Q = \psi'(a)$, and $R = \psi''(a)$,

$$c_0 = 1, \quad c_1 = -P, \quad c_2 = \frac{P^2 - Q}{2}, \quad c_3 = \frac{-P^3 + 3PQ - R}{6}.$$

The standard polygamma–Hurwitz identities can convert these coordinates to zeta values when useful [13, 12].

6.2 The exact entire difference

Theorem 6.1 (Universal singular part under a common shift). *For $\Re a > 0$ and u in a sufficiently small disc centered at zero,*

$$D(s, u; a) := F_a(s, u) - K_a(u) F_1(s, u) \quad (6.7)$$

extends jointly holomorphically to all $s \in \mathbb{C}$, locally in a, u . Consequently, for every $r \geq 0$,

$$D_r(s; a) := M_r(s; a) - \sum_{j=0}^r k_j(a) M_{r-j}(s; 1) \quad (6.8)$$

is entire in s .

For every integer $m \geq 0$, its complete depth-generating value is

$$D(-m, u; a) = - \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} \frac{(a-1)_j}{u+j}. \quad (6.9)$$

Here braces denote Stirling numbers of the second kind and $x_j = x(x-1) \cdots (x-j+1)$. Equivalently,

$$D_r(-m; a) = \Delta_{m,r}(a) := (-1)^{r+1} \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} \frac{(a-1)_j}{j^{r+1}}. \quad (6.10)$$

Thus every value in (6.10) is a rational polynomial in a of degree at most $m+1$.

Proof. In the common half-plane of convergence, the Mellin kernel of F_a is

$$\Gamma(s)F_a(s, u) = \int_0^\infty t^{s-1} G_a(t, u) dt, \quad G_a(t, u) = e^{-at} {}_2F_1(1, a+u; a; e^{-t}). \quad (6.11)$$

Termwise integration follows from absolute convergence. The Gauss connection formula at $z = 1$ [15] simplifies here to

$$\begin{aligned} G_a(t, u) &= K_a(u) \frac{e^{-t}}{(1-e^{-t})^{1+u}} + R_a(t, u), \\ R_a(t, u) &= -\frac{a-1}{1+u} e^{-at} {}_2F_1(1, a+u; 2+u; 1-e^{-t}). \end{aligned} \quad (6.12)$$

Indeed its second local hypergeometric factor is ${}_2F_1(a-1, -u; -u; 1-z) = z^{1-a}$, and its two connection coefficients reduce to $K_a(u)$ and $-(a-1)/(1+u)$. The identity is first read for noninteger u , where the generic connection formula applies, and then continued through $u = 0$. The simplified two terms are individually analytic in u near zero.

The function $R_a(t, u)$ is analytic in t near zero, jointly in the other parameters. Both G_a and the first term in (6.12) decay exponentially as $t \rightarrow +\infty$, locally uniformly for $\Re a > 0$. Thus the same holds for R_a . Write $R_a(t, u) = \sum_{j \geq 0} r_j(a, u) t^j$. For any $L \geq 1$,

$$\begin{aligned} \Gamma(s)D(s, u; a) &= \sum_{j=0}^{L-1} \frac{r_j(a, u)}{s+j} \\ &\quad + \int_0^1 t^{s-1} \left(R_a(t, u) - \sum_{j=0}^{L-1} r_j(a, u) t^j \right) dt + \int_1^\infty t^{s-1} R_a(t, u) dt. \end{aligned}$$

The remainder integrals are jointly holomorphic for $\Re s > -L$. All their parameter derivatives have the same local integrability, possibly with fixed powers of $\log t$. Each displayed simple pole at $s = -j$ is canceled by the simple zero of $1/\Gamma(s)$ there. Increasing L proves joint entireness in s and gives

$$D(-m, u; a) = (-1)^m m! [t^m] R_a(t, u). \quad (6.13)$$

For fixed m , the right side is a polynomial in a of degree at most $m+1$: only the first $m+1$ terms of the local hypergeometric series can contribute, and the j -th term has a factor $(1-e^{-t})^j$. Its coefficients are rational functions of u , analytic near zero.

It remains to identify this polynomial. For a positive integer A , shifting the index of the defining series gives the exact finite identity

$$D(s, u; A) = -K_A(u) \sum_{k=1}^{A-1} \frac{(1+u)_{k-1}}{(k-1)! k^s}.$$

At $s = -m$, use

$$k^m = \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} (k-1)^{\underline{j-1}}.$$

The finite binomial summation yields

$$K_A(u) \sum_{k=1}^{A-1} \frac{(1+u)_{k-1}}{(k-1)!} (k-1)^{\underline{j-1}} = \frac{(A-1)^{\underline{j}}}{u+j}.$$

For completeness, after putting $\ell = k - j$, the nonzero summands are $(1+u)_{j-1}(u+j)_{\ell}/\ell!$; their sum is $(1+u)_{j-1}(u+j+1)_{A-j-1}/(A-j-1)!$. Multiplication by $K_A = (A-1)!/(1+u)_{A-1}$ gives the stated quotient. When $A \leq j$, both sides are zero. Therefore (6.9) holds at infinitely many positive integers A . Both sides are polynomials in a of bounded degree, so they agree identically. Finally expand $(u+j)^{-1} = \sum_{r \geq 0} (-1)^r u^r / j^{r+1}$ to obtain (6.10). $\square$

At $r = 0$, the theorem reduces to the usual Bernoulli-polynomial identity for $\zeta(-m, a) - \zeta(-m)$. At positive depths it gives a finite rational interpolation of the entire difference after the Gamma mixing of depths. The theorem does not identify $M_r(-m; a)$ with a convergent series: the defining series diverges there, and the statement concerns its meromorphic continuation.

6.3 A bridge to independent inner and outer shifts

The same argument unifies the earlier outer-shift calculus with the common-shift family. Let $\Re a > 0$, $\Re b > 0$, and $\Re c > 0$, where $c = 1 + b - a$. Define

$$F_{a,b}(s, u) = \sum_{n \geq 0} \frac{(a+u)_n}{(a)_n (n+b)^s}, \quad B_c(s, u) = \sum_{n \geq 0} \frac{(1+u)_n}{n! (n+c)^s}.$$

The function B_c is precisely the previous generator with an outer shift only. At $a = 1$, $F_{a,b} = B_b$; at $b = a$, $F_{a,b} = F_a$.

Corollary 6.2 (Two shifts and a finite binomial correction). *The difference*

$$D_{a,b}(s, u) = F_{a,b}(s, u) - K_a(u) B_{1+b-a}(s, u)$$

is entire in s , locally in its other parameters. For $m \geq 0$,

$$D_{a,b}(-m, u) = \sum_{v=0}^m \binom{m}{v} (b-a)^{m-v} D(-v, u; a). \quad (6.14)$$

Every term on the right is the explicit finite Stirling sum in (6.9). In depth r the corresponding correction is

$$[u^r] D_{a,b}(-m, u) = \sum_{v=0}^m \binom{m}{v} (b-a)^{m-v} \Delta_{v,r}(a).$$

Proof. Multiplication of the Mellin kernel in (6.12) by $e^{-(b-a)t}$ gives

$$e^{-bt} {}_2F_1(1, a+u; a; e^{-t}) = K_a(u) \frac{e^{-(1+b-a)t}}{(1-e^{-t})^{1+u}} + e^{-(b-a)t} R_a(t, u).$$

The last kernel is analytic at zero and decays exponentially at infinity under the stated hypotheses. The same Mellin subtraction proves entireness. Taking its m -th Taylor coefficient and applying (6.13) proves (6.14) by the ordinary binomial theorem. $\square$

Thus no new singular function is required to separate arbitrary inner and outer shifts in this domain: the existing outer-shift generator carries the singular terms and a fully explicit polynomial correction accounts for the values of the entire remainder at nonpositive integers. The higher spectral derivatives of this remainder remain a separate identity problem.

6.4 The complete Laurent principal part and finite part

Define the rational polynomials $b_j(u)$ and $W_m(u)$ by

$$e^{-t} \left(\frac{t}{1 - e^{-t}} \right)^{1+u} = \sum_{j \geq 0} b_j(u) t^j, \quad (6.15)$$

$$W_m(u) = (u - m)_{m+1} b_{m+1}(u). \quad (6.16)$$

The logarithm of the parenthesized factor is the analytic germ at zero whose value there is zero. Each W_m is a rational polynomial divisible by u . The first three are

$$W_0(u) = \frac{u(u-1)}{2}, \quad (6.17)$$

$$W_1(u) = \frac{u(u-1)(u-2)(3u-1)}{24}, \quad (6.18)$$

$$W_2(u) = \frac{u^2(u-1)^2(u-2)(u-3)}{48}. \quad (6.19)$$

Theorem 6.3 (All depths, every nonpositive integer). *Let $m, r \geq 0$ and $\Re a > 0$. Then*

$$M_r(-m + \varepsilon; a) = \sum_{h=0}^r \frac{[u^{r+1-h}] \{C_a(u) W_m(u)\}}{\varepsilon^h} + \Delta_{m,r}(a) + O(\varepsilon). \quad (6.20)$$

In particular,

$$\text{FP}_{s=-m} M_r(s; a) = [u^{r+1}] \{C_a(u) W_m(u)\} + \Delta_{m,r}(a). \quad (6.21)$$

The finite part belongs to $\mathbb{Q}[a, \psi(a), \psi'(a), \dots, \psi^{(r-1)}(a)]$, where at $r = 0$ the polygamma list is empty.

Proof. For the universal unshifted kernel, Mellin subtraction gives, near $(s, u) = (-m, 0)$,

$$F_1(-m + \varepsilon, u) = \frac{b_{m+1}(u)}{\Gamma(-m + \varepsilon)(\varepsilon - u)} + \frac{Q_m(\varepsilon, u)}{\Gamma(-m + \varepsilon)},$$

where Q_m is jointly holomorphic. The last term contributes only $O(\varepsilon)$ after taking any fixed coefficient in u . Expand $1/\Gamma(-m + \varepsilon)$ in powers of ε ; its constant coefficient is zero. From the first term, the coefficient of $u^r \varepsilon^{-h}$, for $h \geq 0$, is

$$[u^{r+1-h}] \frac{K_a(u) b_{m+1}(u)}{\Gamma(u - m)} = [u^{r+1-h}] \{C_a(u) W_m(u)\}.$$

The second equality is the Gamma recurrence. Finally add the entire function $D(s, u; a)$ and apply (6.10). Since W_m has no constant coefficient, at most the first r derivatives of $\log \Gamma(a + u)$ can occur, giving the stated algebra. $\square$

The polynomial correction in (6.21) is essential. For instance, with $r = m = 0$, omitting it gives $-1/2$, whereas the correct value is $\zeta(0, a) = 1/2 - a$. Thus the singular-coefficient rule for the earlier outer-shift family cannot be transferred to the fully shifted family without its analytic-kernel contribution. This is an extension guard, not a claim that the earlier report made this incorrect transfer.

Corollary 6.4 (Exact pole orders). *At $s = 1$, the order of the pole of M_r is $r + 1$. At $s = 0$ and every negative odd integer $s = -m$, the order is exactly r for $r \geq 1$, with leading coefficient $\zeta(-m)$. At a negative even integer $s = -m$, $m \geq 2$, the order is exactly $r - 1$ for $r \geq 2$, with leading coefficient*

$$\frac{m+1}{2m} B_m. \quad (6.22)$$

At these negative even integers, M_1 is holomorphic. All the specified leading coefficients are independent of a .

Proof. The assertion at one follows from the next theorem. From (6.15),

$$[u]W_m(u) = (-1)^m m! b_{m+1}(0) = \zeta(-m).$$

This is nonzero for $m = 0$ and for positive odd m . If $m \geq 2$ is even, $b_{m+1}(0) = B_{m+1}/(m+1)! = 0$. The two elementary expansions

$$\frac{t}{e^t - 1} = 1 - \frac{t}{2} + \sum_{q \geq 1} \frac{B_{2q}}{(2q)!} t^{2q}, \quad \log \frac{t}{1 - e^{-t}} = \frac{t}{2} - \sum_{q \geq 1} \frac{B_{2q}}{2q(2q)!} t^{2q}$$

give

$$[t^{m+1}] \frac{t}{e^t - 1} \log \frac{t}{1 - e^{-t}} = \frac{m+1}{2m} \frac{B_m}{m!}.$$

Multiplication by the leading term $m!u$ of $(u-m)_{m+1}$ proves $[u^2]W_m = (m+1)B_m/(2m)$. This is nonzero, since every positive even Bernoulli number is nonzero [16]. Thus the zero of W_m at zero is exactly double in this case and simple in the preceding case. Since $C_a(0) = 1$ and D is entire, the pole orders and leading coefficients follow directly from (6.20). $\square$

6.5 The principal Stieltjes point

Theorem 6.5 (Gamma coefficients at the principal pole). *For $r \geq 0$,*

$$M_r(1 + \varepsilon; a) = \sum_{j=0}^r \frac{c_j(a)}{\varepsilon^{r+1-j}} + c_{r+1}(a) + O(\varepsilon). \quad (6.23)$$

Also

$$D(1, u; a) = \frac{K_a(u) - 1}{u}, \quad D_r(1; a) = k_{r+1}(a). \quad (6.24)$$

The apparent singularity on the right is removable.

Proof. Let $p_n = (a+u)_n/(a)_n$. Its exact difference is $p_{n+1} - p_n = up_n/(n+a)$. For small $\Re u < 0$, $p_n \rightarrow 0$; therefore telescoping proves $F_a(1, u) = -1/u$. The leading kernel term in (6.12) shows that

$$F_a(1 + \varepsilon, u) = \frac{C_a(u)}{\varepsilon - u} + Q_a(\varepsilon, u)$$

with Q_a jointly holomorphic near zero. Evaluate at $\varepsilon = 0$ for $\Re u < 0$ to obtain $Q_a(0, u) = (C_a(u) - 1)/u$, and continue holomorphically through $u = 0$. Coefficient extraction proves (6.23). Applying the same telescoping value to the entire difference gives (6.24). $\square$

At depth one, the commonly used inclusive harmonic Hurwitz function is

$$\zeta_H(s, a) = \sum_{n \geq 0} \frac{\sum_{j=0}^n (j+a)^{-1}}{(n+a)^s} = M_1(s; a) + \zeta(s+1, a).$$

Thus (6.23) gives

$$\zeta_H(1 + \varepsilon, a) = \frac{1}{\varepsilon^2} - \frac{\psi(a)}{\varepsilon} + \frac{\psi(a)^2 + \psi'(a)}{2} + O(\varepsilon).$$

The strict and inclusive conventions differ even at the finite part. The formula at all depths gives the Gamma-polygamma coefficients of the strict height-one family; higher regular Laurent coefficients are not claimed to reduce to this same finite algebra.

6.6 Explicit values and Gamma antiderivatives

Writing $P = \psi(a)$ and $Q = \psi'(a)$, the first examples are

$$M_1(\varepsilon; a) = -\frac{1}{2\varepsilon} + a - \frac{1}{2} + \frac{P}{2} + O(\varepsilon), \quad (6.25)$$

$$M_1(-1 + \varepsilon; a) = -\frac{1}{12\varepsilon} + \frac{a^2 + a}{4} - \frac{1}{8} + \frac{P}{12} + O(\varepsilon), \quad (6.26)$$

$$M_1(-2; a) = \frac{8a^3 + 6a^2 - 2a - 3}{72}, \quad (6.27)$$

$$M_2(\varepsilon; a) = -\frac{1}{2\varepsilon^2} + \frac{1+P}{2\varepsilon} + 1 - a - \frac{P}{2} - \frac{P^2 - Q}{4} + O(\varepsilon), \quad (6.28)$$

$$M_2(-1 + \varepsilon; a) = -\frac{1}{12\varepsilon^2} + \frac{3/8 + P/12}{\varepsilon} + \frac{-3a^2 - 15a + 8 - 9P - P^2 + Q}{24} + O(\varepsilon). \quad (6.29)$$

At the half shift, put $L = \gamma + 2 \log 2$, so $\psi(1/2) = -L$ and $\psi'(1/2) = \pi^2/2$. Then

$$M_1(-2; 1/2) = -\frac{1}{48}, \quad (6.30)$$

$$M_2(-2 + \varepsilon; 1/2) = \frac{1}{8\varepsilon} + \frac{L}{8} - \frac{19}{288} + O(\varepsilon). \quad (6.31)$$

In the convergent half-plane, for example, $M_2(s; 1/2)$ is one half the Dirichlet series whose numerator is $(\sum_{j < n} (j+1/2)^{-1})^2 - \sum_{j < n} (j+1/2)^{-2}$. Its displayed pole and finite part follow without a numerical relation search.

There is an exact antiderivative for every depth-one finite part. Put

$$P_m(a) = [u^2]W_m(u) + \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} \frac{(a-1)^j}{j^2} \in \mathbb{Q}[a].$$

Then

$$\text{FP}_{s=-m} M_1(s; a) = P_m(a) - \zeta(-m)\psi(a), \quad (6.32)$$

and on the right half-plane

$$\int \text{FP}_{s=-m} M_1(s; a) da = \mathcal{P}_m(a) - \zeta(-m) \log \Gamma(a) + \text{constant}, \quad \mathcal{P}'_m = P_m. \quad (6.33)$$

For even $m \geq 2$, the Gamma term vanishes and the value in (6.32) is itself a polynomial. For $m = 0, 1$,

$$\begin{aligned} \int \text{FP}_{s=0} M_1(s; a) da &= \frac{a^2 - a}{2} + \frac{1}{2} \log \Gamma(a) + \text{constant}, \\ \int \text{FP}_{s=-1} M_1(s; a) da &= \frac{a^3}{12} + \frac{a^2}{8} - \frac{a}{8} + \frac{1}{12} \log \Gamma(a) + \text{constant}. \end{aligned}$$

Higher derivatives in a consequently give explicit polygamma formulas. This assertion is about finite Laurent coefficients; it does not replace a divergent defining series by termwise integration at a negative order.

6.7 Reproduction and further questions

The accompanying verification script performs 72 floating-point checks at 75 decimal working digits. Its maximum absolute residual is below 8.5×10^{-52} . Thirty checks compare (6.9) with an independent continuation of the original Gamma-ratio Dirichlet series, using its Bernoulli-polynomial asymptotic coefficients and Hurwitz-zeta tails. Two repeats with a larger cutoff and more tail terms have residuals below 1.7×10^{-64} . The remaining checks audit the Gauss connection, its mixed Taylor

coefficients, and (6.24). These are floating-point checks, not interval certificates; the all-parameter proof is the analytic argument above.

A constructive next step is to determine the higher regular Taylor coefficients of the entire difference $D_r(s; a)$ at $s = -m$. Unlike its value, its first derivative includes a genuine integral of the analytic kernel and need not be a rational polynomial. A second direction is to retain independent perturbations of the inner exponents: the finite-product model then requires more than one spectral direction, and the singular part need not be governed by a single Gamma quotient. A third direction is to compare the finite Stirling transforms in (6.10) with established poly-Bernoulli and generalized Bernoulli polynomial systems, with exact normalization matching before claiming an identification. Finally, character-weighted sums in the common shift may yield further finite cyclotomic reductions of the polygamma algebra in (6.21).

7 A weighted arctangent integral in classical trilogarithms

The canonical manuscript, in the paragraph following `cleo:eq:lintanh`, considers the weighted analogue of the Reshetnikov–Lee bilinear integral,

$$K(p, q) = \int_0^{\pi/2} \frac{\arctan(p \sin x) \arctan(q \sin x)}{\sin x} dx, \quad p, q \in \mathbb{R}. \quad (7.1)$$

It explains why the unweighted reduction to one Legendre chi function does not carry over, but gives no finite evaluation for this family. We supply a two-term formula in a single even function built from classical dilogarithms and trilogarithms at real arguments in $(0, 1]$. We also give a normalized ordinary antiderivative and an arbitrary-order Euler integration ladder. The proof of the first result uses parameter differentiation and hyperbolic addition; the original unweighted proof of Lee [19] also uses hyperbolic parameters. The elementary logarithmic reduction used later is part of standard weight-three polylogarithm calculus. Its underlying two-logarithm kernel was treated for real parameters by David H. [18]; the polarization method is not new. The contribution relative to the inspected repository is the explicit evaluation of its weighted two-parameter family, with real domains, normalizations, and branch prescriptions proved. Historical priority for this exact weighted evaluation has not been established.

The integral in (7.1) is ordinary and convergent. Its integrand is $pq x + O(x^3)$ at zero. The principal real arctangent is used throughout. Principal complex logarithms and classical polylogarithms are needed only for the more general ordinary primitive. Their series and integral conventions agree with DLMF §25.12 [17].

7.1 The hyperbolic addition formula

Write

$$\chi_j(z) = \frac{\text{Li}_j(z) - \text{Li}_j(-z)}{2} = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)^j} \quad (|z| < 1)$$

for the Legendre chi function. The values used in the next theorem are real and lie in its ordinary convergent series domain, including the endpoints $\chi_2(1) = \pi^2/8$ and $\chi_3(1) = 7\zeta(3)/8$.

Theorem 7.1 (A real two-term weighted evaluation). *Define the even function*

$$\mathfrak{F}(s) = \frac{\pi^2 |s|}{8} - \frac{7}{4} \zeta(3) + |s| \chi_2(e^{-|s|}) + 2 \chi_3(e^{-|s|}), \quad s \in \mathbb{R}. \quad (7.2)$$

Its value at zero is $\mathfrak{F}(0) = 0$. The apparent lack of smoothness from the absolute values is removable: $\mathfrak{F}$ is real analytic on $\mathbb{R}$, and it is equivalently characterized by

$$\mathfrak{F}''(s) = \frac{s}{2 \sinh s}, \quad \mathfrak{F}''(0) = \frac{1}{2}, \quad \mathfrak{F}(0) = \mathfrak{F}'(0) = 0. \quad (7.3)$$

For every $p, q \in \mathbb{R}$, put $\alpha = \operatorname{arcsinh} p$ and $\beta = \operatorname{arcsinh} q$. Then

$$\boxed{K(p, q) = \mathfrak{F}(\alpha + \beta) - \mathfrak{F}(\alpha - \beta).} \quad (7.4)$$

In particular the formula includes both diagonals $p = \pm q$, either zero parameter, and arbitrary signs without branch changes.

Proof. First, for $s > 0$ use $d\chi_j(e^{-s})/ds = -\chi_{j-1}(e^{-s})$ to obtain

$$\mathfrak{F}'(s) = \frac{\pi^2}{8} - \chi_2(e^{-s}) - s\chi_1(e^{-s}), \quad \mathfrak{F}''(s) = s\chi_0(e^{-s}) = \frac{s}{2 \sinh s}.$$

The limits at zero follow from the endpoint values of χ_2 and χ_3 and from $s\chi_1(e^{-s}) = \frac{s}{2} \log \coth(s/2) \rightarrow 0$. The unique twice-normalized primitive of the real analytic even function $s/(2 \sinh s)$ is real analytic and even. Hence (7.2) agrees with this primitive on the whole real line; this proves all assertions in (7.3).

Differentiate (7.1) once in each parameter. On compact real parameter sets the differentiated integrand is bounded by an integrable function, and therefore

$$\begin{aligned} K_{pq}(p, q) &= \int_0^{\pi/2} \frac{\sin x \, dx}{(1 + p^2 \sin^2 x)(1 + q^2 \sin^2 x)} \\ &= \frac{p \operatorname{arcsinh} p / \sqrt{1 + p^2} - q \operatorname{arcsinh} q / \sqrt{1 + q^2}}{p^2 - q^2} \quad (p^2 \neq q^2). \end{aligned}$$

For the second equality use partial fractions and

$$\int_0^{\pi/2} \frac{\sin x}{1 + p^2 \sin^2 x} \, dx = \frac{\operatorname{arcsinh} p}{p \sqrt{1 + p^2}},$$

whose zero-parameter value is 1 and which follows by $v = \cos x$.

Let $\widetilde{K}(\alpha, \beta) = K(\sinh \alpha, \sinh \beta)$. Multiplication by $\cosh \alpha \cosh \beta$ and the addition formulas for $\sinh$ now give

$$\begin{aligned} \widetilde{K}_{\alpha\beta} &= \frac{\alpha \sinh \alpha \cosh \beta - \beta \sinh \beta \cosh \alpha}{\sinh^2 \alpha - \sinh^2 \beta} \\ &= \frac{1}{2} \left\{ \frac{\alpha + \beta}{\sinh(\alpha + \beta)} + \frac{\alpha - \beta}{\sinh(\alpha - \beta)} \right\}. \end{aligned} \quad (7.5)$$

Both quotients on the last line have their removable value 1 at zero. Thus the equality holds at $\alpha = \pm\beta$ by continuity. By (7.3), this is the mixed derivative of $\mathfrak{F}(\alpha + \beta) - \mathfrak{F}(\alpha - \beta)$. Both this expression and $\widetilde{K}$ vanish on $\alpha = 0$ and $\beta = 0$. Integration of their equal mixed derivatives over the oriented rectangle from $(0, 0)$ to (α, β) proves (7.4) for every real α, β . $\square$

For algebraic real p, q , the two arguments $e^{-|\alpha+\beta|}$ and $e^{-|\alpha-\beta|}$ are algebraic. Indeed $e^\alpha = p + \sqrt{1 + p^2}$ and $e^\beta = q + \sqrt{1 + q^2}$ are positive algebraic numbers, and taking the absolute value in either exponent selects an algebraic number or its reciprocal. Thus (7.4) expresses every such integral by classical polylogarithms at real algebraic arguments and logarithms of positive algebraic numbers. It asserts no independence or minimality of the resulting constants.

The diagonal specializes particularly simply:

$$K(p, p) = \mathfrak{F}(2 \operatorname{arcsinh} p). \quad (7.6)$$

For $\ell = \log(1 + \sqrt{2})$ and $r = 3 - 2\sqrt{2}$, this gives the exact identity

$$\boxed{\int_0^{\pi/2} \frac{\arctan^2(\sin x)}{\sin x} \, dx = \frac{\pi^2 \ell}{4} - \frac{7}{4} \zeta(3) + 2\ell \chi_2(r) + 2\chi_3(r).} \quad (7.7)$$

Its numerical value is 0.718064319741103407036328710693...; the decimal only identifies the normalization, while the theorem proves the identity.

An exact weighted counterpart of the manuscript's Cleo integral also follows. Set $\alpha = \log(1 + \sqrt{2})$ and $\beta = \log((1 + \sqrt{5})/2)$. Then

$$\begin{aligned} \int_0^{\pi/2} \frac{1}{\sin x} \arctan^2\left(\frac{6 \sin x}{3 + \cos 2x}\right) dx \\ = \mathfrak{F}(2\alpha) + 2\mathfrak{F}(\alpha + \beta) - 2\mathfrak{F}(\alpha - \beta) + \mathfrak{F}(2\beta). \end{aligned} \quad (7.8)$$

To verify the reduction put $s = \sin x \in [0, 1]$. Since $\arctan s + \arctan(s/2) < \pi/2$, the tangent addition formula gives

$$\arctan \frac{6 \sin x}{3 + \cos 2x} = \arctan(\sin x) + \arctan\left(\frac{1}{2} \sin x\right).$$

Squaring and integrating yields $K(1, 1) + 2K(1, 1/2) + K(1/2, 1/2)$, proving (7.8) by (7.4).

7.2 The weighted inverse-hyperbolic companion

The same parameter method has a trigonometric form. It gives a second exact family, including convergent improper endpoint values.

Theorem 7.2 (An inverse-hyperbolic bilinear evaluation). *For $|r|, |s| \leq 1$ define*

$$L(r, s) = \int_0^{\pi/2} \frac{\operatorname{arctanh}(r \sin x) \operatorname{arctanh}(s \sin x)}{\sin x} dx,$$

using real inverse hyperbolic tangents and interpreting an endpoint with $|r| = 1$ or $|s| = 1$ as an improper integral. Put

$$\mathfrak{G}(t) = \frac{7}{4}\zeta(3) - t \operatorname{Im} \chi_2(e^{it}) - 2 \operatorname{Re} \chi_3(e^{it}), \quad |t| \leq \pi. \quad (7.9)$$

The chi values on the unit circle are defined by their absolutely convergent series. Then $\mathfrak{G}$ is even, continuous on $[-\pi, \pi]$, and real analytic on $(-\pi, \pi)$, with

$$\mathfrak{G}''(t) = \frac{t}{2 \sin t}, \quad \mathfrak{G}''(0) = \frac{1}{2}, \quad \mathfrak{G}(0) = \mathfrak{G}'(0) = 0. \quad (7.10)$$

For $a = \arcsin r$, $b = \arcsin s$, including either sign of either parameter,

$$\boxed{L(r, s) = \mathfrak{G}(a + b) - \mathfrak{G}(a - b).} \quad (7.11)$$

In particular,

$$\boxed{\int_0^{\pi/2} \frac{\operatorname{arctanh}^2(\sin x)}{\sin x} dx = \frac{7}{2}\zeta(3).} \quad (7.12)$$

Proof. Conjugation of the chi series proves evenness and endpoint continuity of (7.9). For $0 < t < \pi$, polylogarithm differentiation gives

$$\mathfrak{G}'(t) = \operatorname{Im} \chi_2(e^{it}) - t \operatorname{Re} \chi_1(e^{it}), \quad \mathfrak{G}''(t) = t \operatorname{Im} \chi_0(e^{it}) = \frac{t}{2 \sin t}.$$

Here $\operatorname{Re} \chi_1(e^{it}) = \frac{1}{2} \log \cot(t/2)$, so the limit of the first derivative at zero is zero. Also $\mathfrak{G}(0) = 0$ by $\chi_3(1) = 7\zeta(3)/8$. The normalized even primitive of $t/(2 \sin t)$ is real analytic on $(-\pi, \pi)$, proving (7.10).

For $|r|, |s| < 1$, differentiation and partial fractions give

$$L_{rs}(r, s) = \frac{r \arcsin r / \sqrt{1 - r^2} - s \arcsin s / \sqrt{1 - s^2}}{r^2 - s^2} \quad (r^2 \neq s^2).$$

The underlying elementary integral is $\int_0^{\pi/2} \sin x / (1 - r^2 \sin^2 x) dx = \arcsin r / (r\sqrt{1 - r^2})$, again obtained by $v = \cos x$. For $\tilde{L}(a, b) = L(\sin a, \sin b)$ the sine addition formulas therefore yield

$$\tilde{L}_{ab}(a, b) = \frac{1}{2} \left\{ \frac{a + b}{\sin(a + b)} + \frac{a - b}{\sin(a - b)} \right\}, \quad |a|, |b| < \pi/2.$$

The zero quotients take their removable value 1. This is the mixed derivative of the right side of (7.11), and both sides vanish on the two axes. Rectangle integration proves the identity in the open parameter square.

For all $|r|, |s| \leq 1$ the absolute value of the integrand is bounded by $\operatorname{arctanh}^2(\sin x) / \sin x$. This majorant is $O(x)$ at zero and $O(\log^2(\pi/2 - x))$ at the other endpoint, so it is integrable. Dominated convergence extends the equality to the closed square. Finally $\chi_3(-1) = -7\zeta(3)/8$ and $\operatorname{Im} \chi_2(-1) = 0$ give $\mathfrak{G}(\pi) = 7\zeta(3)/2$, proving (7.12). The endpoint value also follows directly from $u = \operatorname{arctanh}(\sin x)$:

$$\int_0^\infty \frac{u^2}{\sinh u} du = 4 \sum_{k=0}^\infty \frac{1}{(2k+1)^3} = \frac{7}{2}\zeta(3).$$

□

For algebraic $r, s \in [-1, 1]$, the chi arguments in this theorem are algebraic points on the unit circle, since $e^{ia} = \sqrt{1 - r^2} + ir$ and $e^{ib} = \sqrt{1 - s^2} + is$. The theorem has been proved directly over real parameters; it does not depend on an unstated continuation of the arctangent formula.

7.3 The classical logarithmic kernel

For $A, B > 0$, we will need only the two types of kernel

$$\mathcal{L}(a, b) = \int_0^1 \frac{\log(1 - at) \log(1 - bt)}{t} dt, \quad (a, b) = (iA, iB) \text{ or } (iA, -iB). \quad (7.13)$$

These integrals have no singularity on their integration path. Define

$$P_c(y) = \log^2 y \log(1 - y/c) + 2 \log y \operatorname{Li}_2(y/c) - 2 \operatorname{Li}_3(y/c). \quad (7.14)$$

The limiting convention is $P_1(1) = -2\zeta(3)$. In each other use of $P_c(1)$ below, $1/c \notin [1, \infty)$, and thus $P_c(1) = -2 \operatorname{Li}_3(1/c)$ directly.

Lemma 7.3 (A finite principal-branch logarithmic reduction). *If $a = iA$, $b = iB$, $A > B > 0$, or if $a = iA$, $b = -iB$, $A, B > 0$, set $r = (1 - a)/(1 - b)$. Then*

$$\boxed{\mathcal{L}(a, b) = \zeta(3) + \operatorname{Li}_3(b/a) + \frac{P_1(1 - a) + P_1(1 - b) - P_1(r) + P_{a/b}(r)}{2}.} \quad (7.15)$$

For the same-sign case with $A < B$, interchange a and b before applying this expression. At the diagonal,

$$\mathcal{L}(a, a) = P_1(1 - a) + 2\zeta(3). \quad (7.16)$$

All quantities in these prescriptions use principal branches; no unstated continuation across a positive-real polylogarithm cut is needed.

Proof. Differentiation using $d \operatorname{Li}_j(z)/dz = \operatorname{Li}_{j-1}(z)/z$ and $\operatorname{Li}_1(z) = -\log(1 - z)$ gives

$$\frac{d}{dy} P_c(y) = \frac{\log^2 y}{y - c}. \quad (7.17)$$

The four extraneous terms cancel in pairs. Consequently

$$\int_0^1 \frac{\log^2(1 - at)}{t} dt = P_1(1 - a) - P_1(1).$$

For the cross term use the polarization identity

$$2 \log(1 - at) \log(1 - bt) = \log^2(1 - at) + \log^2(1 - bt) - \log^2 \frac{1 - at}{1 - bt}.$$

Here the logarithm of the quotient equals the difference of the two principal logarithms on the stated paths. Indeed, in the same-sign case its argument is $-\arctan(At) + \arctan(Bt) \in (-\pi/2, 0)$; in the opposite-sign case it is $-\arctan(At) - \arctan(Bt) \in (-\pi, 0)$.

With $y = (1 - at)/(1 - bt)$ and $c = a/b$, elementary differentiation yields

$$\frac{dt}{t} = \left(\frac{1}{y-1} - \frac{1}{y-c} \right) dy.$$

The quotient-log square therefore has integral

$$P_1(r) - P_1(1) - P_c(r) + P_c(1).$$

Substitution into polarization proves (7.15) and also the diagonal formula.

For completeness, the branch check for the additional polylogarithms is explicit. In the same-sign case $A > B$,

$$y(t) = \frac{1 + ABt^2 + i(B - A)t}{1 + B^2t^2}, \quad \operatorname{Re} \frac{y(t)}{c} = \frac{B/A + B^2t^2}{1 + B^2t^2} < 1.$$

Thus $y(t)$ is in the lower half-plane for $t > 0$, while $y(t)/c$ starts in $(0, 1)$ and avoids $[1, \infty)$. In the opposite-sign case $y(t)$ is in the lower half-plane and $y(t)/c$, with $c < 0$, is in the upper half-plane, starting at $-B/A < 0$. These observations justify every use of (7.17) with the stated principal values. The endpoint $y = 1$ is removable in P_1 because $\log^2 y \log(1 - y) \rightarrow 0$ and $\log y \operatorname{Li}_2(y) \rightarrow 0$. $\square$

Define the real symmetric kernel

$$\mathcal{J}(A, B) = \frac{1}{2} \operatorname{Re} \{ \mathcal{L}(iA, -iB) - \mathcal{L}(iA, iB) \}, \quad A, B > 0. \quad (7.18)$$

Lemma 7.3 makes this a finite expression in classical trilogarithms, dilogarithms, logarithms, and $\zeta(3)$. Conjugation of the logarithms gives the simpler integral meaning

$$\mathcal{J}(A, B) = \int_0^1 \frac{\arctan(At) \arctan(Bt)}{t} dt. \quad (7.19)$$

To see the factor $1/2$, write $\log(1 - iAt) = \frac{1}{2} \log(1 + A^2t^2) - i \arctan(At)$; the real part of the difference in (7.18) is exactly twice the product of arctangents.

7.4 The ordinary primitive at a general endpoint

Theorem 7.4 (Finite classical evaluation and a normalized primitive). *For $p, q \in \mathbb{R}$ put*

$$a = e^{\operatorname{arcsinh} p} = p + \sqrt{1 + p^2} > 0, \quad b = e^{\operatorname{arcsinh} q} = q + \sqrt{1 + q^2} > 0.$$

For $0 < \theta < \pi$ let $T = \tan(\theta/2)$. Then

$$\begin{aligned} & \int_0^\theta \frac{\arctan(p \sin x) \arctan(q \sin x)}{\sin x} dx \\ &= \mathcal{J}(aT, bT) - \mathcal{J}(aT, T/b) - \mathcal{J}(T/a, bT) + \mathcal{J}(T/a, T/b). \end{aligned} \quad (7.20)$$

In particular,

$$\boxed{K(p, q) = \mathcal{J}(a, b) - \mathcal{J}(a, b^{-1}) - \mathcal{J}(a^{-1}, b) + \mathcal{J}(a^{-1}, b^{-1})}. \quad (7.21)$$

Together with Lemma 7.3, these are finite formulas using only classical polylogarithms of orders at most three. They include negative and zero values of p, q and the diagonals $p = \pm q$.

Proof. The half-angle substitution $t = \tan(x/2)$ gives

$$\sin x = \frac{2t}{1+t^2}, \quad \frac{dx}{\sin x} = \frac{dt}{t}.$$

Since $a - a^{-1} = 2p$,

$$\arctan \frac{2pt}{1+t^2} = \arctan(at) - \arctan(t/a), \quad t > 0. \quad (7.22)$$

The tangents of the two sides coincide. Both sides lie strictly between $-\pi/2$ and $\pi/2$ because the two arctangents on the right belong to $(0, \pi/2)$. Hence the equality has no additive multiple of π , including when $p < 0$. Apply (7.22) for p and q , expand the product, and rescale each integral over $[0, T]$ to $[0, 1]$. Equation (7.19) gives (7.20). The choice $\theta = \pi/2$ gives $T = 1$ and proves (7.21). $\square$

The right side of (7.20), plus an arbitrary constant, is an ordinary antiderivative in θ on $(0, \pi)$. Its constant has been fixed by the limit zero at $\theta = 0$. For algebraic p, q and algebraic $T > 0$, every argument in the finite formula is algebraic. This includes rational multiples θ/π with $0 < \theta < \pi$. The statement is an explicit upper bound on the kinds of functions needed for the expression; it asserts no independence or minimality of its special values.

7.5 An elementary differential certificate

The elementary computation in the proof of Theorem 7.1 supplies a differential certificate for both finite formulas, including their diagonal limits.

Proposition 7.5 (Mixed derivative). *For real p, q with $p^2 \neq q^2$,*

$$\partial_p \partial_q K(p, q) = \frac{p \operatorname{arcsinh} p / \sqrt{1+p^2} - q \operatorname{arcsinh} q / \sqrt{1+q^2}}{p^2 - q^2}. \quad (7.23)$$

At $p^2 = q^2$ the continuous value, for $p \neq 0$, is

$$\frac{1}{2(1+p^2)} + \frac{\operatorname{arcsinh} p}{2p(1+p^2)^{3/2}}, \quad (7.24)$$

and the value at $(0, 0)$ is 1.

Proof. Equation (7.23) was proved above. Take its divided-difference limit by differentiating the numerator as a function of p^2 to obtain (7.24). Its continuous limit at zero is 1. $\square$

Since $K(p, 0) = K(0, q) = 0$, its mixed derivative determines K uniquely by integration over the rectangle with corners $(0, 0)$ and (p, q) . This supplies an elementary differential certificate for any further compressed trilogarithm representation of the same family.

7.6 Every Euler integration order

The half-angle reduction is not confined to weight three. For integers $m \geq 1$ set

$$\mathcal{L}_m(a, b) = \frac{1}{(m-1)!} \int_0^1 \frac{(-\log t)^{m-1} \log(1-at) \log(1-bt)}{t} dt. \quad (7.25)$$

For the imaginary arguments used below this is an ordinary convergent integral. The convention for double polylogarithms is

$$\operatorname{Li}_{r,1}(x, y) = \sum_{n>k \geq 1} \frac{x^n y^k}{n^r k}.$$

Theorem 7.6 (A depth-two Euler primitive ladder). *For $m \geq 1$,*

$$\mathcal{L}_m(a, b) = \text{Li}_{m+1,1}(a, b/a) + \text{Li}_{m+1,1}(b, a/b). \quad (7.26)$$

This is initially a series identity for $|a|, |b| < 1$ and nonzero a, b . For purely imaginary nonzero a, b , continue the two arguments simultaneously along $(a, b) \mapsto (\lambda a, \lambda b)$, $0 < \lambda \leq 1$. This radial prescription gives the value in (7.25); no singularity is crossed.

Define

$$\mathcal{J}_m(A, B) = \frac{1}{2} \text{Re}\{\mathcal{L}_m(iA, -iB) - \mathcal{L}_m(iA, iB)\}, \quad A, B > 0,$$

and abbreviate

$$D_p(t) = \arctan(at) - \arctan(t/a) = \arctan \frac{2pt}{1+t^2}.$$

Set

$$\begin{aligned} \mathcal{I}_m(T; p, q) &= \mathcal{J}_m(aT, bT) - \mathcal{J}_m(aT, T/b) \\ &\quad - \mathcal{J}_m(T/a, bT) + \mathcal{J}_m(T/a, T/b). \end{aligned} \quad (7.27)$$

Then

$$\mathcal{I}_m(T; p, q) = \frac{1}{(m-1)!} \int_0^T \log^{m-1}(T/t) D_p(t) D_q(t) \frac{dt}{t}. \quad (7.28)$$

Writing $\Theta_T = T d/dT$,

$$\Theta_T \mathcal{I}_m = \mathcal{I}_{m-1} \quad (m \geq 2), \quad \Theta_T \mathcal{I}_1 = D_p(T) D_q(T). \quad (7.29)$$

Thus every fixed integration order has a finite expression in double polylogarithms of weight $m+2$ and depth at most two.

Proof. Expand the logarithms for $|a|, |b| < 1$ and integrate term by term. Absolute convergence gives

$$\mathcal{L}_m(a, b) = \sum_{j,k \geq 1} \frac{a^j b^k}{jk(j+k)^m}.$$

Use $1/(jk) = (1/j + 1/k)/(j+k)$ and write $n = j+k$. The two resulting sums are exactly those in (7.26). Along the indicated imaginary radial paths, $1 - \lambda at$ and $1 - \lambda bt$ never vanish for $0 \leq t \leq 1$; analytic continuation gives the asserted extension. The real-part calculation establishing (7.19) is unchanged by the logarithmic weight. The four-term expansion and the scaling $t = Ts$ prove (7.28). Differentiating its normalized Cauchy integration kernel gives (7.29). Near $t = 0$ the product $D_p(t) D_q(t)$ is $O(t^2)$, which justifies every differentiation and fixes the primitive constants. $\square$

One useful diagonal identity follows without another integration:

$$\mathcal{J}_m(A, A) = 2^{-m-1} \text{Li}_{m+1,1}(-A^2, 1) - 2 \text{Re} \text{Li}_{m+1,1}(iA, 1). \quad (7.30)$$

Indeed, polarization of $\log(1-at) + \log(1+at) = \log(1-a^2t^2)$ gives

$$\mathcal{L}_m(a, -a) = 2^{-m} \text{Li}_{m+1,1}(a^2, 1) - \text{Li}_{m+1,1}(a, 1) - \text{Li}_{m+1,1}(-a, 1),$$

where $t \mapsto t^2$ accounts for 2^{-m} . Combine this with $\mathcal{L}_m(a, a) = 2 \text{Li}_{m+1,1}(a, 1)$. The arguments in (7.30) lie off the positive-real cut, so the same radial continuation is unambiguous.

7.7 An exact obstruction to the one-product ansatz

The earlier paragraph's failure of the unweighted factorization can be made into a very short exact statement. Set

$$u(p) = \frac{p}{1 + \sqrt{1 + p^2}}, \quad u(0) = 0.$$

The unweighted integral is a function of $u(p)u(q)$. The weighted integral has no such analytic factorization, even into an unspecified one-variable function.

Proposition 7.7. *There is no function G analytic near zero for which $K(p, q) = G(u(p)u(q))$ holds for all sufficiently small real p, q .*

Proof. For $|p|, |q| < 1$, integration of the absolutely convergent arctangent series gives

$$K(p, q) = \sum_{j, k \geq 0} \frac{(-1)^{j+k} p^{2j+1} q^{2k+1}}{(2j+1)(2k+1)} \frac{4^{j+k}}{(2j+2k+1) \binom{2j+2k}{j+k}}. \quad (7.31)$$

In particular,

$$K(p, q) = pq - \frac{2}{9}(p^3q + pq^3) + \text{terms of total degree at least six},$$

whereas

$$u(p)u(q) = \frac{pq}{4} - \frac{p^3q + pq^3}{16} + \text{terms of total degree at least six}.$$

The coefficient of pq would force $G'(0) = 4$, and hence the coefficient of p^3q would have to be $-1/4$, contradicting $-2/9$. Terms of order at least two in G cannot contain p^3q . $\square$

This is an obstruction only to the stated functional ansatz. It says nothing about independence of particular constants. Theorem 7.1 shows explicitly that a short formula does exist after changing to the hyperbolic sum and difference of the parameters. Unlike a bounded search for one χ_3 , the coefficient mismatch proves the exact assertion for every analytic G .

Reproduction and next questions. The accompanying verification program checks (7.17) and the nonfactorization coefficients symbolically, compares both classical formulas with direct real quadrature, verifies the hyperbolic mixed derivative and the ordinary primitive at several endpoints, tests the inverse-hyperbolic family at signed and nearly saturated endpoints, and checks the higher integration orders against independent nested series in a common convergence disk. These calculations audit normalizations and branch choices; the proofs above establish the identities. Natural next tasks are to identify additional endpoints at which the four-kernel ordinary primitive compresses to real one-variable kernels, to seek analogous addition formulas for higher Euler integration orders, and to determine which specializations of that higher-weight ladder admit classical polylogarithm reductions. The Gaussian S_6 and current S_8 candidates remain unresolved by this argument.

8 Source audit, reproducibility, and further research

8.1 Corrections and exact claim status

Two narrow issues remain in the pinned canonical `chapters/03-algebraic.tex`. Both had been flagged in an earlier continuation and are confirmed here; they are not presented as newly discovered mistakes. The proposed patch preserves the existing labels and the proved identities.

The real logarithm in `cleo:eq:lintanh`. The formula concerns real $|\lambda| < 1$, with $\theta = \arcsin \lambda$. Its term $\theta \log \tan(\theta/2)$ must be

$$\theta \log \left| \tan \frac{\theta}{2} \right|,$$

with continuous value zero at $\theta = 0$, unless the parameter range is restricted to positive λ . For negative λ the displayed logarithm otherwise introduces an imaginary term into a real integral. For example, at $\lambda = -1/2$ the spurious principal-logarithm term is $-i\pi^2/6$. The surrounding Clausen terms are unchanged. The real identity is checked by differentiation in θ , giving $\theta/\sin \theta$ on both sides, and the zero value at the origin.

The description of the unweighted diagonal. The phrase “the simplest non-elementary diagonal” is replaced by “the $(1, 1)$ diagonal has the Legendre–chi representation.” The exact formula

$$\int_0^{\pi/2} \arctan^2(\sin x) dx = \pi \chi_2(3 - 2\sqrt{2})$$

is retained. A representation alone does not prove minimality or nonreduction in a specified algebra of constants. This correction does not concern the different, weighted integral evaluated in (7.7).

The package contains a unified patch and its before/after hashes. The patch passed a dry-run application and was applied to a temporary copy, whose bytes matched the intended correction. The baseline itself was unchanged.

Source item	Status after this report
<code>gauss:eq:S4-closed</code>	Already proved in the canonical <code>04-S4-proof.tex</code> ; this report does not claim a new proof of its rational octahedral certificate.
<code>cycloquot:conj:S6</code>	The short S_6 identity remains conjectural. The existing incomplete-beta and reflection generators do not supply the missing odd-weight relation.
<code>s8new:conj:S8</code>	The current S_8 candidate remains conjectural. It is distinct from the rejected earlier vector recorded under <code>research:prop:S8-rejected</code> .
Higher golden vectors	The numerical Li_6 – Li_9 candidates in <code>03-algebraic.tex</code> have not acquired proofs here.
Colliding unequal grids	Resolved for every pair of Stieltjes indices and argument orders by Section 3, including the full subtraction polynomial and convergent collision matching.
Nonlinear local coordinates	Resolved for orientation-preserving analytic local diffeomorphisms by Section 4, including the composition law and finite-jet dependence.
Three unequal frequencies	Resolved under pairwise grid separation by Section 5, including all argument and order coefficients. Colliding triples remain a separate problem.
Fully shifted height-one sums	An all-depth extension, with the depth-one literature credited. The inner shift must not be confused with the older outer-only shift or with simultaneous changes of all exponents.
Weighted arctangent family	Evaluated for all real parameters, with ordinary and Euler primitives and a direct inverse-hyperbolic companion. No global priority or minimality claim is made.

The separated-grid theorem in the incoming report is valid under its stated separation hypothesis. Extending its distributional product directly through collision would be an invalid use of that theorem, rather than an error in the separated statement. Similarly, omitting $\Delta_{m,r}(a)$ from the fully

shifted harmonic formula would give an incorrect extension; the present report does not attribute that omitted correction to an existing source. These distinctions matter when the new sections are integrated.

8.2 What the computational artifacts establish

The analytic proofs establish the parameter-uniform identities. The computer checks target signs, scales, coordinate constants, finite coefficient extraction, and independent numerical evaluations. They do not infer identities from small residuals. All numerical results are ordinary floating-point audits; no result file is described as an interval certificate.

Family	Independent audit	Observed residual
Nonlinear coordinates	213 exact rational-series checks, including cutoff constants and composition; 4 subtracted integrals at 70 digits	Absolute $< 4.0 \cdot 10^{-36}$
Colliding products	33 exact coefficient checks; 9 polygamma integrals at 44 digits and 2 Stieltjes derivative integrals at 36 digits	Scaled $< 6.7 \cdot 10^{-45}$ and $< 2.8 \cdot 10^{-31}$
Fully shifted sums	72 checks at 75 digits, using Gamma-ratio Dirichlet tails and direct local differentiation	Absolute $< 8.5 \cdot 10^{-52}$
Three frequencies	2 exact rational Bernoulli integrals compared with 12 weighted-kernel Mellin integrals at 60 digits	Absolute $< 7.2 \cdot 10^{-66}$
Weighted primitives	78 comparisons at 80 digits; symbolic checks of the mixed derivatives, primitive, and coefficient obstruction	Absolute $< 1.3 \cdot 10^{-79}$

There are 167 numerical comparisons in total. The difference between working precision and observed residual is expected: some checks use finite Taylor or asymptotic tails, while others compare small exact values with direct quadrature. The results record the actual precision and method rather than a uniform advertised number of correct digits. For the harmonic continuation, two larger-tail repeats give residuals below $1.7 \cdot 10^{-64}$, which tests the identified truncation risk.

The seven mathematical scripts are independent of the TeX build. They use Python, SymPy, and mpmath. A driver reruns them, writes their logs, and validates the result files. The exact collision generator also accepts m, n, r, k as command-line arguments and prints the corresponding finite symbolic identity, with formal symbols for ordinary zeta derivatives. The package includes the source manifest, a claim ledger, integration instructions, the correction patch, and both modular and standalone TeX. No numerical relation search is needed to reproduce any proof in this report.

8.3 Proposed exact research problems

The following questions concern identities and structural reductions. They are questions, not additional theorems. Each starts from a specific formula proved above.

- Q1. Mixed collisions of three unequal grids.** Remove pairwise separation in Theorem 5.2. Classify pairwise collision points and triple collision points, write their complete local subtraction polynomials, and determine whether the regular constant in every simultaneous collision limit agrees with the merged ambient finite part. The two-grid result suggests a positive matching statement for suitably specified collision coordinates, but multiple rates can introduce distinct logarithmic terms and must be tracked explicitly.
- Q2. An all-factor finite congruence formula.** For $\ell \geq 4$ pairwise separated integer grids, generalize the six sign-cone proof to a finite collection of colored multiple Tornheim or Mordell–Tornheim kernels on the rank- $(\ell - 1)$ lattice $\sum p_j n_j = 0$. Determine a useful minimal finite character formula, then give every ambient-coordinate Stieltjes coefficient with complete local

numerators. The challenge is an exact finite organization of the lattice, rather than existence of meromorphic continuation alone.

- Q3. Which odd-order cancellations survive at more than two factors?** Corollary 3.3 cancels an entire antisymmetric two-factor spectral kernel. Identify a permutation or character projection for three or more factors that removes the genuinely multiple spectral terms and leaves only local Stieltjes data. A useful first case has equal Stieltjes indices and one total argument derivative.
- Q4. Nonlinear transport of merged products.** Theorem 4.1 applies to the full singular expansion of a colliding product. Compose it with (3.6) to obtain a single closed multivariate generator for every nonlinear pullback of every two-grid product. Determine sharp vanishing thresholds in the two Stieltjes indices. For a nonunit tangent, factor out the affine contribution before classifying the curvature terms.
- Q5. A finite algebra for coordinate composition.** Express the coefficients of (4.4) and (4.7) in a consistently normalized Bell-polynomial basis and identify the resulting composition matrices. Which logarithmic singular germs have zero defect for every coordinate tangent to the identity? The theorem supplies a direct coefficient test, but a classification by singular order and logarithmic degree could expose additional exact cancellations.
- Q6. Higher regular Laurent coefficients of the harmonic entire difference.** Evaluate $\partial_s^k D(-m, u; a)$ for $k \geq 1$. Its value at $k = 0$ is a finite Stirling sum, whereas differentiation introduces logarithmic Mellin moments of the analytic Gauss block. Seek closed expressions in Gamma derivatives, generalized Stieltjes constants, and a precisely specified class of multiple polylogarithms. Do not assume the rational polynomial closure at $k = 0$ persists for $k \geq 1$.
- Q7. Identify the Stirling polynomial system exactly.** Compare $\sum_j \left\{ \begin{smallmatrix} m+1 \\ j \end{smallmatrix} \right\} (a-1)^j / j^{r+1}$ with poly-Bernoulli and related polynomial systems, including the falling-factorial convention and the shift by one. An exact generating function identification could yield addition, reflection, and differentiation identities for the complete harmonic correction.
- Q8. Independent inner exponents.** Replace the inner exponent string $(1, \dots, 1)$ in M_r by independent parameters, keeping the common shift. Determine the smallest generating kernel whose singular part still separates by a Gamma factor, and identify the extra entire corrections. This would connect the present height-one calculus to the incoming all-exponent nested jets without conflating their spectral directions.
- Q9. Rational shifts and character projections.** Apply finite Dirichlet-character sums to the a -dependence of the Laurent formulas. Determine exact reductions of their Gamma-polygamma coefficients to Dirichlet L -values and their derivatives, and identify which character parities eliminate the polynomial Stirling correction. The finite correction should be summed before imposing parity.
- Q10. A real addition law for higher Euler primitives.** The first weighted arctangent integral has the two-term formula (7.4); the higher Euler primitives currently have the finite depth-two expression (7.27). Determine which higher integration orders admit a comparable addition formula in $\alpha \pm \beta$, and which require irreducible two-variable functions. The known exact coefficient obstruction concerns one product variable only; it does not rule out these other organizations.
- Q11. Relations among the algebraic endpoint values.** The weighted Cleo identity combines logarithms of the silver and golden units with χ_2 and χ_3 at explicit algebraic arguments. Seek functional-equation certificates for further reductions at these arguments. A target basis should be specified in advance, and failure of a finite relation search should not be interpreted as independence.
- Q12. The short S_6 and current S_8 candidates.** Extend the convergent octahedral or lifted functional-equation mechanism that proved S_4 to weights seven and nine. The existing reflection projection cancels at the relevant odd total weights, so a successful proof needs an additional

functional relation. A useful deliverable would be an exact rational row certificate with an independently checked analytic origin for every row. The frozen candidate vectors, not newly fitted alternatives, should remain the targets.

Q13. Formalization of the local finite-part lemmas. Formalize the elementary Taylor-subtraction coefficient lemma first, including its exact test-function and coordinate conventions. It can then serve as a common foundation for nonlinear contacts and colliding products. The finite rational coefficient computations are suitable for checked algebraic certificates, while the analytic continuation and dominated-differentiation steps remain separate proof obligations.

The most immediate continuations are Q1, Q4, and Q6: their inputs are explicitly available in this report, and each asks for an exact identity whose first unresolved term can be computed without a numerical basis guess. The S_6 and S_8 problems remain more structurally demanding; the present endpoint calculus supplies normalization discipline, not the missing weight-seven or weight-nine functional equation.

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